AI-ENABLEDSMARTLAMPPOSTFORREALTIME EMERGENCY DETECTION, LOCALIZATION, AND MULTI-CHANNEL ALERTING

APROJECTREPORT

Submittedby

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BONAFIDE CERTIFICATE

Certified that this Project report titled "**AI-ENABLED SMART LAMP POST FOR REAL-TIME EMERGENCY DETECTION, LOCALIZATION, AND MULTI-CHANNEL ALERTING**" is the bonafide work of "**KAVIYA V (2116231901020)**" who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Safety systems deployed across densely populated urban and suburban regions predominantly operate in a response-oriented manner: an emergency usually comes to the attention of authorities only after a bystander observes it, locates a phone, and places a call. This project report presents SingaPen AI, a crisis-managementsystemintegrateddirectlyintostreetlightpolesthatcombines physicalsensing,on-devicevisualprocessing,andmultipleindependentwireless communication channels into a single economical unit. An MFRC522 RFID module paired with an ESP32 microcontroller allows an individual to trigger a linked identity alert, while a manual SOS button provides a parallel human-triggered pathway. Built-in redundancy is provided through a SIM800L GSM module and an SX1278 LoRa transceiver operating at 868 MHz, so that alert delivery does not depend on any single network. An onboard camera running UltralyticsYOLO11nperformsedge-basedweapondetectionasanadditionalAI verificationlayer.AllincomingdataisroutedtoaFlaskbackend,withaFirebase Realtime Database collecting the live notification feed and a Leaflet-based dashboard mapping real-time incidents for dispatchers. The system accommodatestwohardwareformsastationarySmartRFIDEmergencyPoleand acompactPortableGuardianSOSdevicebothsharingthesamealertschemaand cloud backend. Campus deployment and testing at Rajalakshmi Institute of Technology, Chennai, measured sub-second end-to-end alert latency over WiFi (803ms),reliableLoRadeliveryovershortrange(100%packetsuccessat1km, 85%at5km),anddependableweapon-classAIdetectionwitha5%falsepositive rate. These results indicate that integrating AI-powered emergency detection directly into existing streetlight infrastructure is an economical and effective approach to improving public safety response times.

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TABLEOFCONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**ii**
	LISTOFTABLES	**vi**
	LISTOFFIGURES	**vii**
1	**INTRODUCTION**	1
	BACKGROUND	
	PROBLEMSTATEMENT	
	OBJECTIVES	
	SCOPE	
	SECURITYREQUIREMENTS/SECURITY CONSIDERATIONS	
	SDGALIGNMENT	
	COMPARISONOFCONVENTIONALAND PROPOSED RESPONSE	
	ORGANIZATIONOFTHEREPORT	
2	**LITERATUREREVIEW**	7
	EXISTINGSYSTEMWITHSECURITY APPROACHES	
	RESEARCHGAPS	
	PROPOSEDSYSTEM	
	OVERVIEWOFTHEPROPOSEDSYSTEM	
	OBJECTIVESOFTHEPROPOSEDSYSTEM	
	ADVANTAGESOFPROPOSEDSYSTEM	
3	**SYSTEMDESIGN**	14
	SYSTEMARCHITECTURE	
	MODULEDESIGN	
	DATAFLOW/UMLDIAGRAMS	
	THREATMODELANDATTACKSURFACE	
	SECURITYCONTROLS	
	LAYEREDVIEWOFTHEARCHITECTURE	
	ALERTDATAMODEL	
	TRACEABILITYOFOBJECTIVESTODESIGN	

4 SYSTEMDESCRIPTION 26
DATACOLLECTION
INSTALLATIONLAYOUTOFTHESMART RFID EMERGENCY POLE
ACQUISITIONANDTIMINGPARAMETERS
DATAPREPROCESSING
ALGORITHMS/METHODOLOGY
RESULTSANDDISCUSSION
GRAPHICALANALYSISOFRESULTS
TESTSCENARIOSANDOBSERVED OUTCOMES
DEPLOYMENTCOSTPROJECTION
LIMITATIONSANDMITIGATION
5 CONCLUSION 41
KEYFINDINGSANDCONTRIBUTIONS
ROADMAPFORFUTUREENHANCEMENTS
REFERENCES 44
APPENDIX 46

LISTOFTABLES

TABLE NO.	TITLE	PAGE NO.
1.1	SDGALIGNMENT	4
1.2	CONVENTIONALEMERGENCYREPORTINGVERSUSSINGAPEN AI	6
2.1	COMPARISONWITHREPRESENTATIVEEXISTING SYSTEMS	9
2.2	COMPARISONOFTHETHREEALERTCHANNELS USEDINSINGAPENAI	10
3.1	COREHARDWAREMODULES–SMARTRFID EMERGENCYPOLE	16
3.2	THREATANALYSISANDMITIGATION	22
3.3	SECURITYCONTROLSBYCOMMUNICATION CHANNEL	23
3.4	LOGICALFIELDSOFTHEALERTRECORD	25
3.5	TRACEABILITYOFPROJECTOBJECTIVESTO DESIGNFEATURES	25
4.1	KEYPERFORMANCEMETRICS	27
4.2	ACQUISITIONANDOPERATINGPARAMETERS	30
4.3	TESTSCENARIOSANDOBSERVEDOUTCOMES	38
4.4	ESTIMATEDCOSTPERKILOMETREOFCORRIDOR	38
4.5	OBSERVEDLIMITATIONSANDMITIGATION	39
5.1	FUTUREENHANCEMENTS,BENEFITSAND CHALLENGES	41

LISTOFFIGURES

FIGURE NO	TITLE	PAGE NO.
1.1	CONVENTIONALEMERGENCYREPORTINGVERSUS THESINGAPENAIRESPONSEFLOW	5
2.1	POSITIONINGOFSINGAPENAIINTHEREVIEWED RESEARCHLANDSCAPE	12
3.1	SINGAPENAISYSTEMARCHITECTURE	14
3.2	HARDWAREBLOCKDIAGRAMOFTHESINGAPENAI SYSTEM	15
3.3	SEQUENCEDIAGRAMFORRFIDTRIGGER	17
3.4	USECASEDIAGRAMOFTHESINGAPENAISYSTEM	18
3.5	STATEDIAGRAMOFTHEESP32FIRMWARE	19
3.6	FLOWCHARTOFALERTDETECTIONANDMULTI-CHANNELTRANSMISSION	20
3.7	LEVEL-1DATAFLOWDIAGRAMOFTHESYSTEM	21
3.8	ATTACKSURFACEOFTHEALERTPATHWITH THREATMARKERST1–T7	22
3.9	LAYEREDVIEWOFTHESINGAPENAI ARCHITECTURE	24
4.1	PORTABLEGUARDIANSOSDEVICE	27
4.2	COMPONENTARRANGEMENTONTHESMARTRFID EMERGENCYPOLE	28
4.3	FRONTVIEWOFSINGAPENAIPROTOTYPEDEVICE	29
4.4	BACKSIDEVIEWOFSINGAPENAIPROTOTYPE	29
4.5	EDGEVISIONPIPELINEFORWEAPON-CLASS VERIFICATION	32
4.6	SCHEMATICLAYOUTOFTHEDISPATCHER DASHBOARD	33
4.7	AI-BASEDWEAPONDETECTIONALERTTRIGGERED AUTOMATICALLY WITH LOCATION	34
4.8	SOSALERTTRIGGERANDDASHBOARDINDICATOR WITHLOCATION	34
4.9	END-TO-ENDALERTLATENCYBYCHANNEL	37
4.10	DELIVERYSUCCESSOFTHELORAANDGSM CHANNELS	37
5.1	SUGGESTEDPHASINGOFFUTUREENHANCEMENTS	42

SDGALIGNMENT

S. No.	SDG No. &Goal	RelevantSDGTarget	Project Contribution/ Alignment
1	SDG5 – Gender Equality	5.2 – Eliminate violence againstwomenandgirls in publicandprivatespaces	Portable Guardian SOSdeviceand RFIDEmergency Poleprovidealow-friction panic triggerwithsub-secondalert deliverytoroute incidentstothe nearestresponse unit, including all womenpolice stations.
2	SDG11 – Sustainable Citiesand Communities	11.7 – Provide universal accesstosafe,inclusive and accessiblepublicspaces	Embedding identity, AI threat verification,and multi-channel alertinginto existing streetlight infrastructure improvesthesafety ofpublicspacesata per-unitcost realisticfor municipaland campusbudgets.

CHAPTER 1

INTRODUCTION

BACKGROUND

In urban areas, safety incidents frequently go unnoticed, and this delays emergency response.Thisespecially affectswomen,students, andelderlyresidentsnavigatingpoorly lit or isolated stretches of public space. Today, a victim or witness must locate a phone, recall an emergency contact number, place a call, and then state their exact location an entire sequence of manual steps that itself accounts for the response delay. When no bystander is present, or the victim is physically unable to act, the process breaks down completely.

PassiveinfrastructuresuchasconventionalCCTVnetworksarguablycompoundsthe problem rather than solving it, since footage is recorded and archived for later review, offering little value during the incident itself. SingaPen AI was conceived specifically to close this intervention gap by folding emergency identification and alerting directly into the lampposts already present on nearly every street corner. Each pole-mounted unit integrates an RFID identity scanner, a manual SOS button, GPS tracking, and a camera performing AI-based threat verification. Once triggered, the system dispatches a verified alerttoacommandcenterwithinroughlyonesecond,carryingGPScoordinatesand,where applicable, an AI-flagged threat image.

Three field observations shaped the design of the system. First, dependence on a single wireless channel is risky, since both cellular networks and WiFi tend to degrade undercrowddensity,andlosingthatonechannelmeanslosingthealertentirely.Second,a passive camera with no human trigger cannot register subjective fear a person may sense danger well before any weapon becomes visible, and an object-detection model has no conceptofthatfear.Third,costdictatesdeployability:intheIndiancontext,per-unitprice is arguably the largest obstacle to scaling this kind of hardware, so the system had to be assembledfromcommodity,off-the-shelfcomponentscostingonlyafewthousandrupees per unit.

PROBLEMSTATEMENT

Most existing public safety devices optimize for a single dimension of the problem either communication reach, as with GSM/LoRa trackers, passive monitoring, as with CCTVnetworkslackinganypanicmechanism,oraccesscontrolalone,aswithstandalone RFID systems. Rarely do these approaches combine all three with fault-tolerant transmission, and almost none are priced for large-scale municipal rollout.

A genuinely deployable system must satisfy four requirements simultaneously: it needs a low-friction physical trigger that lets someone raise an alarm instantly; where a camera is present, it must automatically verify the threat using AI rather than simply recordingit;itmusttransmitthealertacrossmultipleindependentnetworkssothatasingle network error does not stop the broadcast; and it must remain affordable enough that covering every 50 to 100 metres along a transit corridor is financially realistic.

OBJECTIVES

- Design and build a functional lamp-post-integrated emergency pole combining RFID identification, a manual SOS trigger, GPS tracking, and YOLO-based AI verification.
- DevelopacompanionportableSOSdevicethatsharestheidenticalalertschemasothat it can accompany an individual anywhere.
- Implementathree-channeltransmissionlayer(WiFi/Firebase,LoRa,GSM)thatdegrades gracefully rather than failing outright when a network drops.
- Build a Flask-based command center dashboard capable of mapping incidents in real time.
- Empirically evaluate the hardware for latency, reliability, and AI detection accuracy under realistic operating conditions.

SCOPE

Thescopeofthisprojectcoversthedesign,prototyping,andcampus-levelevaluation of both hardware forms of SingaPen AI the fixed Smart RFID Emergency Pole and the PortableGuardianSOSdevicealongwiththesupportingcloudbackend(FlaskRESTAPI, Firebase Realtime Database) and the Leaflet-based dispatcher dashboard. The work includes the RFID identity pathway, manual SOS trigger, GPS-based localization, the on-deviceYOLO11nweapon-detectionpipeline,andtheredundantWiFi/LoRa/GSM

transmission layer, together with empirical measurement of latency, packet delivery ratio, detection accuracy, and batterylife. Theproject is limitedto asingle-hop LoRalink and a prototypebillofmaterialsvalidatedthroughacampusdeploymentatRajalakshmiInstitute of Technology, Chennai; large-scale municipal rollout, mesh/relay LoRa topologies for extended range, and regulatory certification of the hardware are identified as future scope rather than part of the current deliverable.

SECURITYREQUIREMENTS/SECURITYCONSIDERATIONS

Several specific threats were considered during design. Alert spoofing is mitigated through per-device authentication tokens verified by the Flask API. RFID cards are inherently easy to clone, so the card is treated only as a hint of identity rather than a full authentication factor, with UIDs cross-checked server-side. WiFi routers were assumed likely to fail or face jamming, motivating the LoRa and GSM fallback paths. To guard against GPS interception over the air, the Firebase link is TLS encrypted, and LoRa payloadsarekeptminimalenoughthatlittleofvaluecouldbeextractedevenifintercepted.

TrafficcrossingtheWiFinetworkisencryptedunderTLS1.2orlaterviatheFirebase SDK. Stored alerts and profiles in the cloud are protected by Firebase security rules, restrictingaccesstoauthenticateddispatchersonly.BecauseLoRaandGSMoperateunder tight bandwidth constraints, the full payload is not encrypted on those channels; instead, onlyabaredeviceIDandcoordinatepairistransmitted,withtheserverresolvingcomplete incidentdetailsinternallyoncetheinitialsignalarrives.Becausethesystemtrackslocation and captures visual data from public spaces, deployment requires clear signage indicating an AI-monitored area, strict data-retention limits to avoid indefinitely storing footage of ordinary daily activity, dashboard access restricted to authorized dispatchers only, and informed consent for any RFID-linked profile data, limited strictly to the minimum information required to assist someone during an emergency.

SDGALIGNMENT

The project aligns with United Nations Sustainable Development Goal 5 (Gender Equality) and Goal 11 (Sustainable Cities and Communities), and its subscription-free, open design was chosen deliberately to keep the system accessible regardless of socio-economic status.

TABLE1.1SDGALIGNMENT

S. No.	SDGNo.&Goal	RelevantSDG Target	Project Contribution/ Alignment
1	SDG5–Gender Equality	5.2– Eliminate violenceagainst womenandgirlsin publicand private spaces	PortableGuardian SOSdeviceand RFIDEmergency Poleprovidealow-frictionpanictrigger withsub-second alert deliverytoroute incidentstothe nearestresponseunit, includingall-women police stations.
2	SDG11– SustainableCities and Communities	11.7 – Provide universalaccessto safe,inclusiveand accessiblepublic spaces	Embeddingidentity, AIthreatverification, andmulti-channel alertingintoexisting streetlight infrastructure improvesthesafety ofpublicspacesata per-unitcostrealistic formunicipaland campusbudgets.

COMPARISONOFCONVENTIONALANDPROPOSEDRESPONSE

ToappreciatetheinterventiongapthatSingaPenAIaddresses,itisusefultocompare the sequence of actions a person must complete in a conventional emergency report with the sequence required by the proposed system. In the conventional flow, shown in Fig. 1.1(a), the victim must first recognize the threat, retrieve and unlock a phone, dial an emergencynumber,waitforthecalltobeanswered,anddescribeboththeincidentanda

locationthatmaynotbeknownprecisely.Eachstepdependsonthevictimremainingcalm and physically able to act, and each one adds delay before a responder is even informed.

In the proposed flow, shown in Fig. 1.1(b), the same person performs a single deliberateaction:tappinganRFIDcardonapoleorpressingtheSOSbuttonontheportable device. Location capture, payload construction, multi-channel transmission, and map display are then carried out automatically, and the camera on the pole can independently raise an AI-corroborated alert without any human input at all.

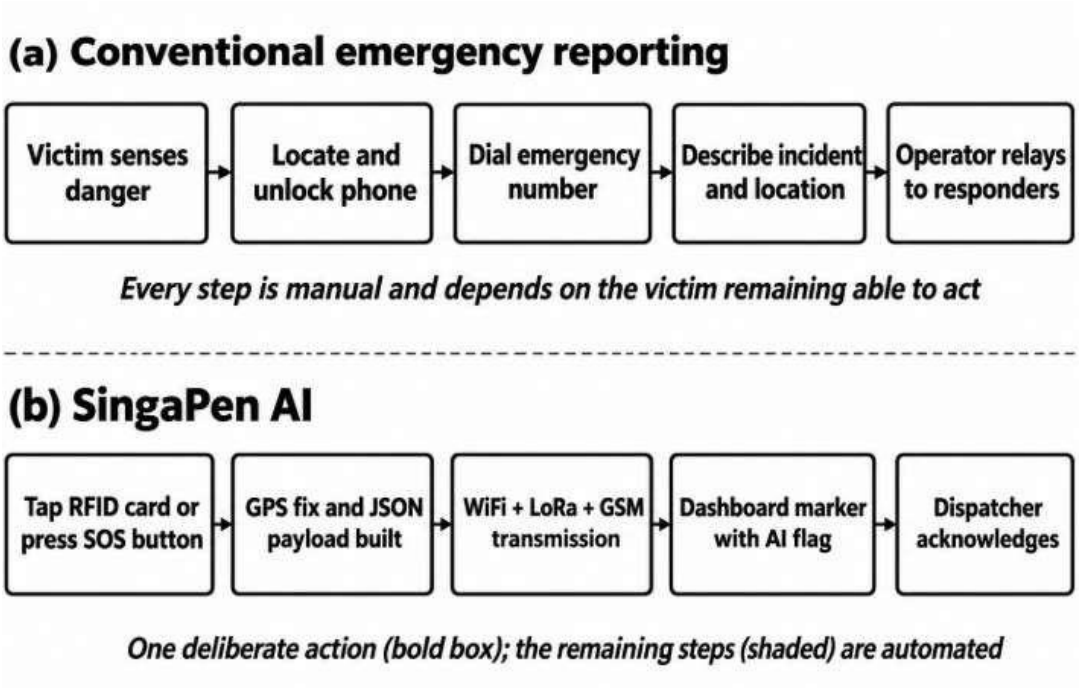


Fig.1.1ConventionalEmergencyReportingversustheSingaPenAIResponse Flow

Table 1.2 contrasts the two approaches across the dimensions that matter most to a dispatcher: the effort demanded of the victim, how location and identity are established, whether the threat is verified, how many networks the alert can use, and how long ittakes toreachthecontrolroom.ValuesquotedforSingaPenAIarethosemeasuredinthecampus evaluation reported in Chapter 4.

TABLE1.2CONVENTIONALEMERGENCYREPORTINGVERSUSSINGAPENAI

Aspect	Conventionalcall	SingaPenAI
Triggereffort	Unlockphone,dialanumber,waitfor the call to connect	A single RFID tap or SOS press (3-secondlongpressontheportableunit)
Location	Describedverballybythecaller	GPScoordinatesattachedautomatically (2.8 m CEP)
Identity	Dependsonthecaller	RFID-linkedprofileretrievedfrom Firebase, with consent
Threatverification	Noneuntilaresponderarrives	YOLO11n weapon-class flag (confidence≥0.20),reviewedbythe dispatcher
Network dependence	Asinglecellularcall	Threeindependentchannels: WiFi/Firebase, LoRa, GSM
Timetoreach control room	Dependsonthecallerandthecall queue	≈803ms(WiFi),≈1,253ms(LoRa), 6–8 s (GSM SMS)
Requirementon victim	Mustbeabletospeakandact	Minimalaction;thecameracanraisean alert without any button press

ORGANIZATIONOFTHEREPORT

The remainder of this report is organized into four further chapters, a list of references, and an appendix.

- Chapter2(LiteratureReview)surveys wearable andpole-mountedsafetydevices,AI-based weapon detection, LoRa emergency communication, and smart-pole research, identifies the gaps in existing work, and introduces the proposed system.
- Chapter3 (System Design)presents thesystem architecture, moduledesign, data flow andUMLdiagrams,threatmodel,securitycontrols,layeredview,alertdatamodel,and the traceability of design features to project objectives.
- Chapter 4 (System Description) describes data collection on the pole and the portable device,thevisionpipelineandmathematicalformulation,andtheexperimentalresults, together with test scenarios, a cost projection, and limitations.
- Chapter 5 (Conclusion) summarizes the key findings and sets out a roadmap of future enhancements.
- The Appendix covers applications, cost and maintenance, ethical considerations, a sample alert payload and a commissioning checklist.

CHAPTER 2

LITERATUREREVIEW

EXISTINGSYSTEMWITHSECURITYAPPROACHES

WearableandPole-MountedPersonalSafetyDevices

Wearable and pole-mounted personal safety systems have been studied extensively. AcommondesignpatterncouplesaGPSreceiverwithaGSMmodulesothatadevicecan transmit its location as a text message [1]. Many implementations rely on an ESP32 or Arduino board together with a NEO-6M GPS unit and a SIM800-series GSM module, sendingcoordinatestoapredefinedcontactlistuponabuttonpress.Someresearchershave pursued hands-free triggering through biometric inputs such as heart-rate monitors [1], where a sudden spike is used as the SOS trigger. Other prototypes combine a physical button with GSM, GPS, and WiFi to notify both personal contacts and local police [2]. A related approach pairs an ESP32 camera with motion and acoustic sensors to capture images automatically during a distress event [3]. Several survey papers catalogue these GPS- and GSM-based combinations in depth [4]. Another design direction employs combinedchipsetssuchastheSIM808,handlingGSMandGPStogetherandoftenpairing themwithaccelerometerstodetectphysicalaltercationswithoutrequiringtheusertopress anything [5]. A recurring limitation across this body of work, however, is reliance on a single communication path, leaving these devices vulnerable to a single point of failure.

AI-BasedWeaponDetectionSystems

A second body of relevant work concerns automated weapon detection through neuralnetworks.TheYOLO(YouOnlyLookOnce)architecturereframedobjectdetection as a single-pass inference task, making real-time video analysis practical for the first time [6].SuccessiveversionsfromYOLOv3throughYOLOv8havesincebeendeployedacross IoT surveillance networks [7]. Some teams have successfully run YOLOv4 pipelines on constrainededgehardware[8],whileotherstrainedYOLOv5specificallyoncustomknife andfirearmdatasets[9].ComparativestudiesofYOLOv7andYOLOv8reportmeasurable gains in mean average precision for weapon classes [10]. Additional work has focused on compressingYOLOvariantsforseverelyresource-limitedhardware[11].Aseparateline

of inquiry moves away from visible-spectrum cameras entirely, exploring thermal and millimeter-waveimagingforconcealedweapondetection[12],[13],whichremainslargely orthogonal to the visible-spectrum camera used in this project but is a promising area for future extension.

LoRa-BasedEmergencyCommunication

LoRaiswidelyfavoredforemergencycommunicationowingtoitslowpowerdraw and independence from cellular infrastructure. Some projects have built self-powered, hand-cranked LoRa devices with integrated GPS for disaster-response scenarios [14]. Others have assembled LoRa mesh networks from inexpensive ESP32 and SX1262/SX1278 hardware, demonstrating self-healing behavior without requiring a centralLoRaWANgateway.Trigger-basedconfigurationshavealsobeenexplored,where a microcontroller transmits a LoRa signal directly to a receiver station that raises an alert on receipt, proving that reliable single-hop delivery is achievable without any internet dependency [15]. Solar-powered LoRa mesh deployments extend this further, sustaining operation entirely off-grid. Collectively, this literature supports the choice of LoRa as a dependable backup channel alongside GSM and WiFi.

SmartPoleandStreet-LightIntegration

A final relevant thread concerns smart-pole research, which demonstrates that embedding electronics and sensors into street-lighting infrastructure is practically viable. Existing systems perform adaptive lighting control and fault detection, while others incorporateapanicbuttonpairedwithacloud-connectedcamera.Somedesignsgofurther, convertingthepoleintoabroaderutilityhubequippedwithsolarpanelsandenvironmental sensors.ThisprojectextendsthatlineofworkbyintroducingRFID-basedidentitytracking together with on-device AI threat verification, a combination that, to the best of available knowledge, had not previously appeared in street-lighting platforms.

TABLE2.1COMPARISONWITHREPRESENTATIVEEXISTING SYSTEMS

SystemClass	**Comm. Channels**	**AIVerif.**	**RFID ID**	**Approx. Cost**
GPS/GSM wearable[2]	GSMonly	No	No	Low
Biometric wearable[1]	GSM	No	No	Moderate
CCTV+panic pole	WiFi/cloud	No	No	Moderate
YOLO surveillance[9]	Network	Yes	No	Mod-High
LoRadisaster node [15]	LoRaonly	No	No	Low
Smartutility pole	WiFi/cloud	No	No	High
SingaPenAI (proposed)	WiFi+LoRa+ GSM	Yes	Yes	Low

CommunicationTechnologiesforEmergencyAlerting

The choice of radio technology largely determines whether an emergency alert arrivesatall.WiFioffershighthroughputandlowlatencybutislimitedtothecoverageof anaccesspointanddependsontherouteranditsbackhaulremainingoperational.Cellular GSMofferswidecoveragethroughexistinginfrastructureandworkswithabareSIMcard, butSMSdeliveryisstore-and-forwardanditslatencydependsoncarrierload,asthe6–8s deliverytimesinthisworkconfirm.LoRa,achirp-spread-spectrummodulationusedinthe sub-GHzISMbands,tradesdatarateforrangeandreceiversensitivity,whichmakesitwell suited to short alert frames such as a device identifier and a coordinate pair. Table 2.2 summarizesthepropertiesofthethreechannelsusedbySingaPenAI,withmeasuredvalues from Chapter 4 where available.

TABLE2.2COMPARISONOFTHETHREEALERTCHANNELSUSED IN SINGAPEN AI

Channel	Roleinthesystem	Coverage	Datarate	Infrastructure dependence	Measured latency
WiFi / Firebase	Primary route; carries the full payload including profileandAIflag	Access-point range(tensof metres)	High (Mbps class)	Routerand internet backhaul	≈803ms
LoRa 868 MHz	Cellular-independent backupusinga minimalframe	Kilometres in line of sight (100%at1km, 85%at5km measured)	Low(kbps class)	Own receiver only;nocarrier network	≈1,253ms
GSMSMS	Last-resort notification to savedemergency contacts	Cellularcoverage area	Verylow; 160 characters per SMS	Carriernetwork	6–8 s (7,303 msaverage)

Because the three channels fail for different reasons (router outage or jamming for WiFi,rangeandlineofsightforLoRa,carriercongestionforGSM),transmittingonallof themsimultaneouslymakesthealertpathconsiderablymorerobustthananysingle-channel design in the surveyed literature.

Privacy,EthicsandGovernanceofPublic-SpaceSensing

Placing cameras and identity readers in public spaces raises legitimate privacy and governancequestions,andasafetysystemshouldaddressthembydesignratherthanasan afterthought. Widely accepted baseline expectations include data minimization, purpose limitation,informedconsentforanypersonalprofile,definedretentionlimitsforimagery, androle-basedaccessforoperators.InIndia,theDigitalPersonalDataProtectionAct,2023 formalizes obligations around consent and purpose for personal data, which makes such design choices a compliance matter as well as an ethical one. Most of the reviewed prototypes concentrate on technical feasibility and say little about these safeguards. SingaPen AI therefore treats signage, retention limits, minimal RFID profiles, and dispatcher-only dashboard access as explicit design requirements (Section 1.5 and Appendix A.3).

RESEARCHGAPS

- Existing wearable and pole-mounted safety devices predominantly rely on a single communication path (GSM-only or LoRa-only), leaving them vulnerable to a single point of failure when that network degrades or fails under crowd density.
- AI-basedweapon-detectionliteratureislargelydeployedaspassivenetwork-connected surveillance, without a parallel human-triggerable pathway that can register a threat before it becomes visually apparent to a camera.
- Smart-pole and street-light integration research demonstrates lighting control, fault detection, and basic panic buttons, but does not combine RFID-based identity linkage with on-device AI verification and redundant multi-channel alerting in a single unit.
- Very little of the reviewed literature reports approximate per-unit cost, making it difficult to judge whether these designs are realistic for large-scale municipal or campus deployment; SingaPen AI addresses this by keeping the bill of materials withinafewthousandrupeesperunitwhilestillofferingthreeindependentchannels and AI corroboration.

Fig. 2.1 places the reviewed work in three overlapping areas: multi-channel alert communication, AI visual verification, and identity or human-triggered alerting. The reviewed systems typically occupy one of these areas, or at most two, whereas SingaPen AI is designed to combine all three in a single low-cost unit.

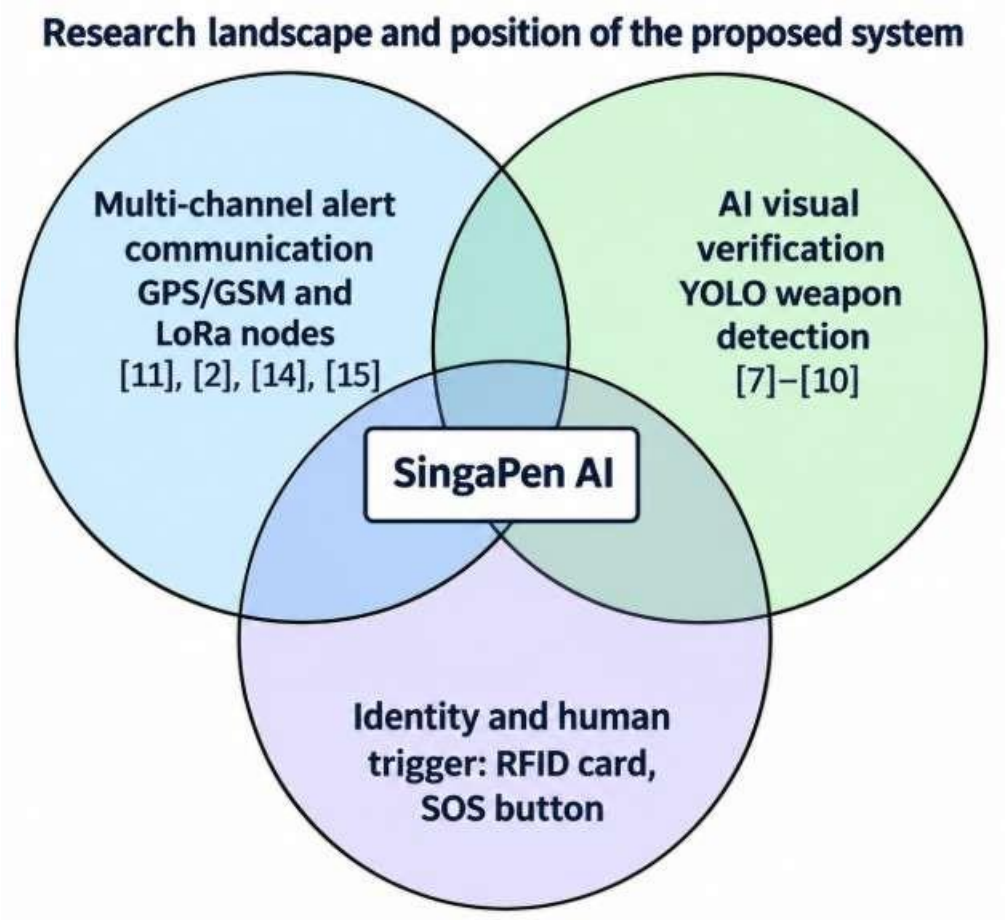


Fig.2.1PositioningofSingaPenAIintheReviewedResearchLandscape

PROPOSEDSYSTEM

OverviewoftheProposedSystem

SingaPenAIiscomposedofthreeinterconnectedparts.TheSmartRFIDEmergency Pole forms the fixed hardware layer, housing the microcontroller, RFID reader, camera, andradiomodules.ThePortableGuardianSOSDeviceisabattery-poweredwearablebuilt aroundthesamedataschema.Thecloudlayer,backedbyFirebaseRealtimeDatabaseand aFlaskapplication,powerstheLeaflet-basedcommandcentermapthattiesbothhardware forms together. A human-triggerable RFID or SOS pathway runs in parallel to the AI verification layer, and transfer happens across three fully independent channels — WiFi/Firebase, LoRa, and GSM — so that no single network outage can silence an alert.

ObjectivesoftheProposedSystem

The proposed system aims to: (1) pair RFID-based identity with AI visual confirmationsodispatchersreceiveapreliminary threatassessmentratherthanabaremap

marker; (2) provide a redundant three-channel transmission system with empirically measured delivery ratio and latency; (3) unify the alert schema and cloud backend across both hardware forms, so alerts from a fixed pole or a portable wearable converge on the same dashboard; and (4) keep the bill of materials remarkably low — the portable device costing approximately Rs. 2,125 — while preserving full multi-channel redundancy.

AdvantagesofProposedSystem

- RelativetosingleGPS/GSMwearables,SingaPenAIofferstwofully independent fallback channels (LoRa and GSM) alongside an automated AI corroboration step that most comparable designs omit.
- ComparedwithstandardYOLO-basedsurveillancesetups,theadditionofaphysical, human-triggerable SOS pathway combined with RFID identity linkage represents a meaningful improvement, since a person may sense danger well before any weapon becomes visible to a camera.
- Theredundantchanneldesigndegradesgracefully:WiFi/Firebaseremainsthefastest primary route (≈ 803 ms), LoRa provides a dependable cellular-independent alternative (≈ 1,253 ms), and GSM SMS, though slower (6–8 s), still reaches emergency contacts when both other options are inaccessible.
- Thebillofmaterialskeepsunitcostwithinreachofmunicipalandcampusbudgets, with no recurring subscription fees since the design uses standard GSM billing, the Firebase free tier, and open-source firmware.

CHAPTER 3

SYSTEMDESIGN

SYSTEMARCHITECTURE

The pole unit is built around an ESP32 dual-core microcontroller functioning as the central controller. An MFRC522 module handles 13.56 MHz RFID cards, allowing identity-linked emergency triggers. A NEO-6M GPS receiver delivers location information,whileaSIM800LGSMmodulesendsSMSnotificationsandanSX1278LoRa transceiver operating at 868 MHz provides a long-range backup communication channel. A camera module delivers frames for AI processing, and power is controlled via a buck converter that ensures a consistent 5V/3.3V power rail. All incoming data is directed to a Flask backend; Firebase Realtime Database collects the live notification feed, while a Leaflet-based map shows real-time incidents for dispatchers.

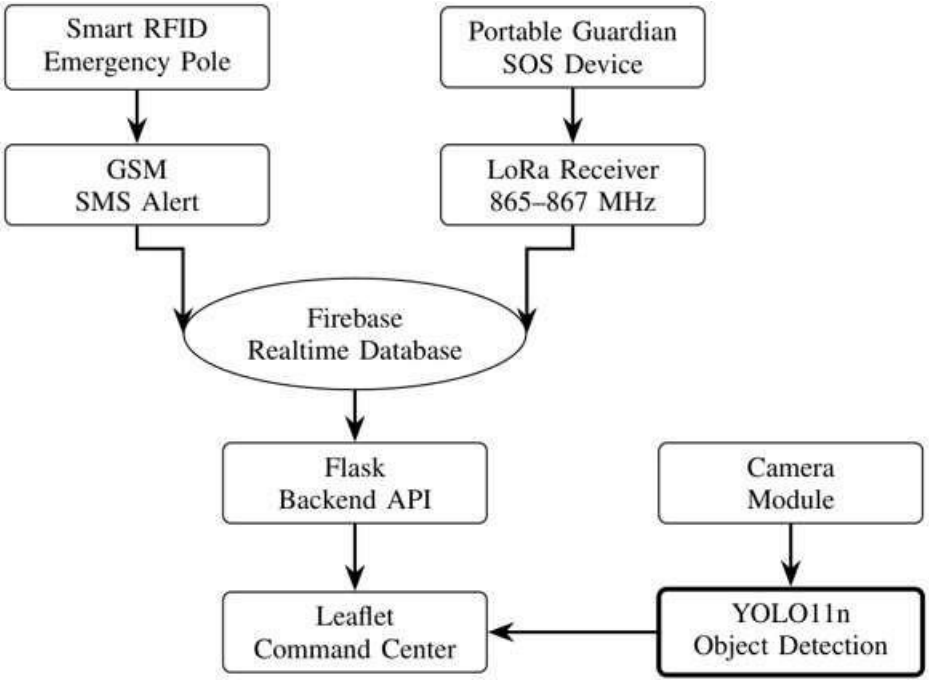


Fig.3.1SingaPenAISystemArchitecture

MODULEDESIGN

TheESP32firmwarerunsanon-blockingeventloopthatcontinuouslypollstheRFID reader, monitors the SOS GPIO interrupt, and serializes outgoing alerts as JSON. On the server side, a Flask application provides REST endpoints to receive incoming alerts, process dispatcher acknowledgments, and serve dashboard queries. Firebase Realtime Database was selected specifically to deliver low-latency updates to linked clients. The dashboard is a standalone Leaflet.js application that connects to Firebase and displays incident markers, real-time camera thumbnails, and AI threat badges directly on the map.

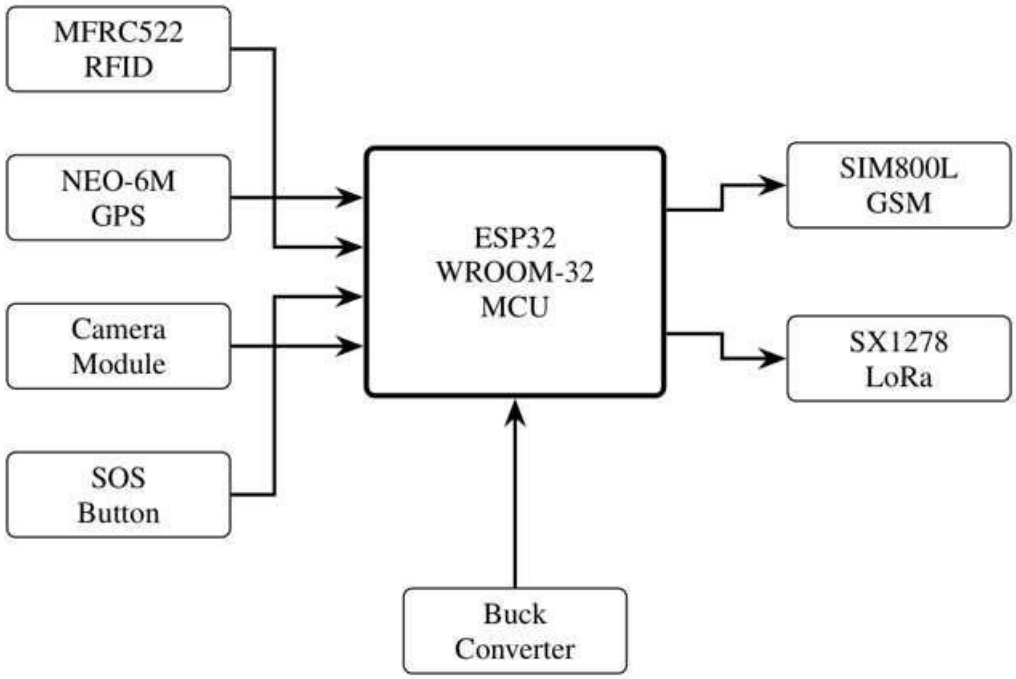


Fig.3.2HardwareBlockDiagram oftheSingaPenAI System

TABLE3.1COREHARDWAREMODULES–SMARTRFIDEMERGENCY POLE

Module	**Function**	**Interface**
ESP32	Centralcontroller	-
MFRC522	RFIDidentification	SPI
NEO-6M	GPSgeolocation	UART
SIM800L	GSMSMS alerting	UART
SX1278	LoRa868MHzlink	SPI
Cameramodule	ImageacquisitionforAI	Serial/CSI
Buckconverter	Powerregulation	-
Speaker/mic	Audio confirmation	I2S/analog
SOSbutton	Manualtrigger	GPIOinterrupt

FlaskRESTAPI

- POST/api/alert –allowshardwaretopushanewJSON payload.
- GET/api/alerts/active–letsthedashboard retrieveunresolvedincidents.
- POST/api/alert/{id}/ack–handlesdispatcher acknowledgmentofa case.
- GET/api/device/{id}/status –providesabasichealthcheck.

FirebaseDatabaseSchema

- devices/{device_id}–tracksregistrationandheartbeatstatus.
- alerts/{alert_id}–storestimestamps,GPScoordinates,triggertype,andtheAI verification flag.
- users/{uid}–holdsRFID-linkedprofilessodispatcherscanidentifywhorequires assistance.

DATAFLOW/UMLDIAGRAMS

TheRFIDtriggerflow,showninFig.3.3,isstraightforward:apersontapstheircard, theESP32readstheUID,thecorrespondingprofileisretrievedfromFirebase,andthe

resultingJSONalertispushedtoFirebase,whichinturnplacesamarkeronthedispatcher dashboard.

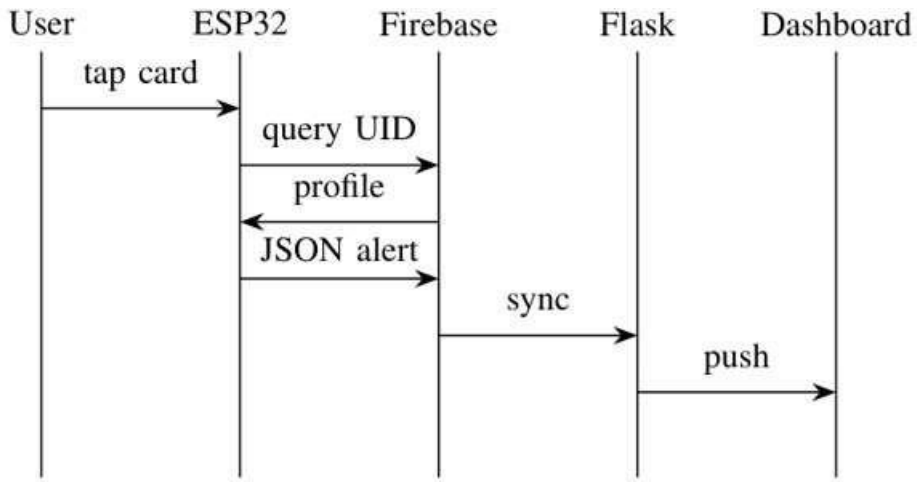


Fig.3.3SequenceDiagramforRFID Trigger

Threeactorsaredefined withinthesystem:theCivilian,theAIVerificationEngine, and the Dispatcher. Civilians raise alerts through the card reader or the SOS button, while the AI engine can independently flag a threat. The dispatcher monitors the dashboard, acknowledgingandresolvingincidentsastheyarrive.Thestandardactivityflowproceeds as:

→Trigger→Validate→CollectGPS→ConstructPayload→Transmit(across three channels simultaneously) → Cloud Update → Dashboard Notify → Dispatcher Response→IncidentResolution.

TheESP32firmwarecyclesthroughfourstates.ItbeginsinStandby,pollingsensors while conserving power. Upon detecting a trigger, it transitions to Triggered to handle debouncingandauthentication,thenmovestoTransmittingtobroadcasttheJSONpayload acrossallradios.ItfinallyentersConfirmed,awaitingacknowledgmentoratimeout,before returning to Standby.

UseCaseDiagram

Fig. 3.4 shows the use cases supported by the system and the actors that take part in them.TheCivilianraisesalertsbytappinganRFIDcardorpressingtheSOSbutton,while the AIVerification Engine independently verifies weapon classes in camera frames. Each ofthesetriggersincludesthecommonstepoftransmittingthealertovertheavailable

channels. The Dispatcher, the only actor with access to the dashboard, views incidents on the map and acknowledges and resolves them.

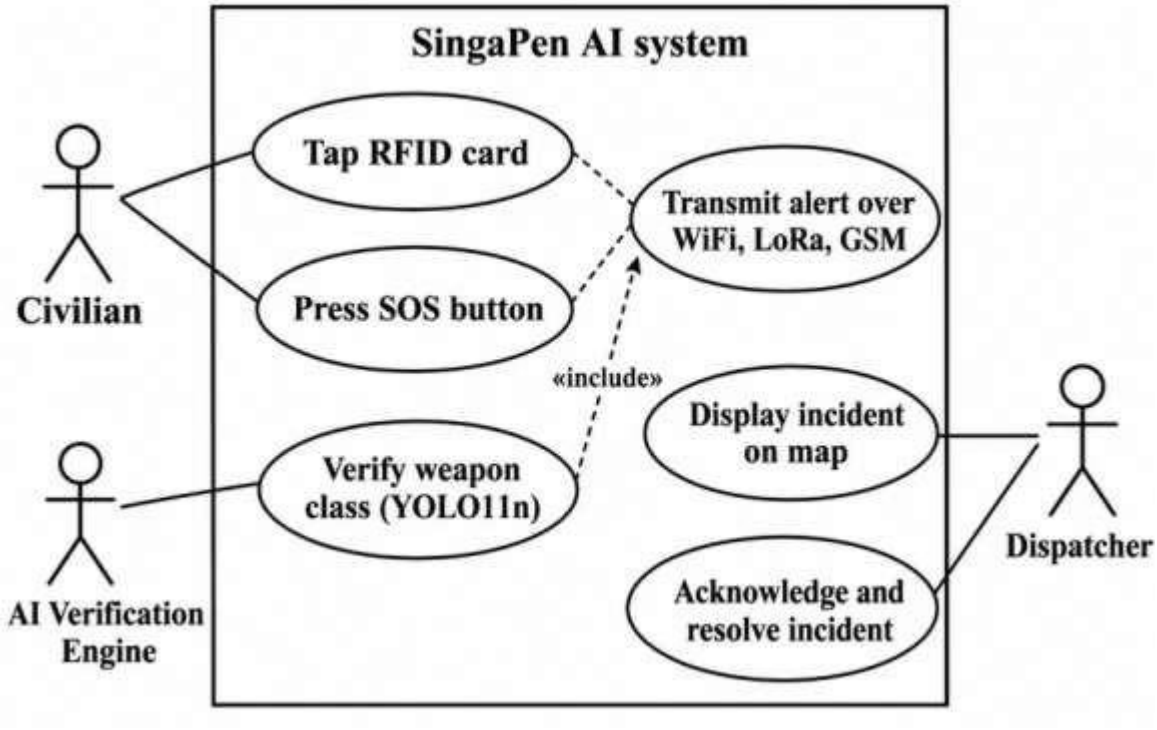


Fig.3.4UseCaseDiagram of theSingaPen AISystem

FirmwareStateDiagram

Thefourfirmwarestatesdescribedabovearesummarizedin Fig.3.5. Standbyisthe default state, in which the device conserves power while polling the RFID reader and the SOS interrupt. A valid trigger moves the firmware to Triggered, where the input is debounced and authenticated; a contact bounce or an unreadable card returns it directly to Standby. In Transmitting, the JSON payload is sent on all radios, and in Confirmed the device waits for an acknowledgement or a timeout before returning to Standby. Keeping the state machine this small makes the behavior predictable and easy to verify.

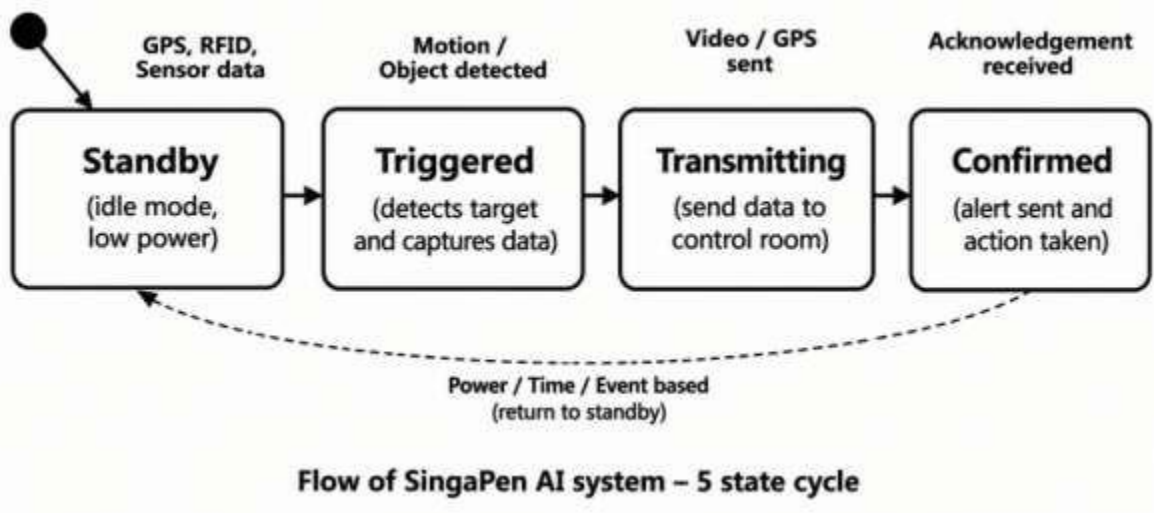


Fig.3.5StateDiagram of theESP32 Firmware

AlertProcessingFlowchart

Fig.3.6expandstheactivityflowdescribedearlierintoaflowchart.Afteratriggeris detected and validated, the device obtains a GPS fix (using the most recent cached fix on the portable unit to save time), assembles the JSON payload, and transmits it on the WiFi/Firebase, LoRa, and GSM paths in parallel. Because the paths are independent, the loss of any one ofthem does not prevent the alertfrom reaching the control room through another.Thecloudupdatecreatesthedashboardmarker,andthedevicereturnstoStandby once the dispatcher acknowledges the alert or the timeout expires.

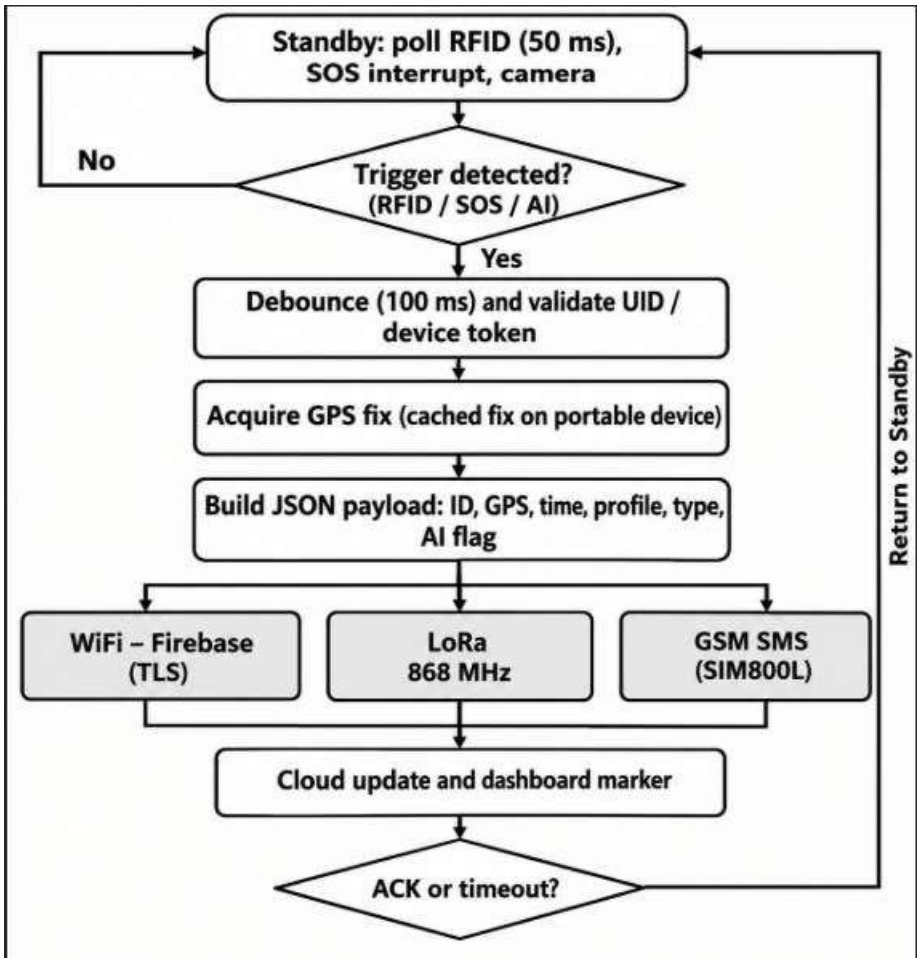


Fig.3.6FlowchartofAlertDetectionandMulti-ChannelTransmission

DataFlowDiagram

Themovementofdatabetweenexternalentities,processes,anddatastoresisshown as a Level-1 data flow diagram in Fig. 3.7. Trigger events from the civilian (process 1.0) andAIflagsfromthecamerapipeline(process2.0)convergeonprocess3.0,whichbuilds andtransmitsthealert.Process4.0storesalertsandservesthemtothedispatcherinterface (process 5.0), while the users store supports profile lookup for RFID-linked identities.

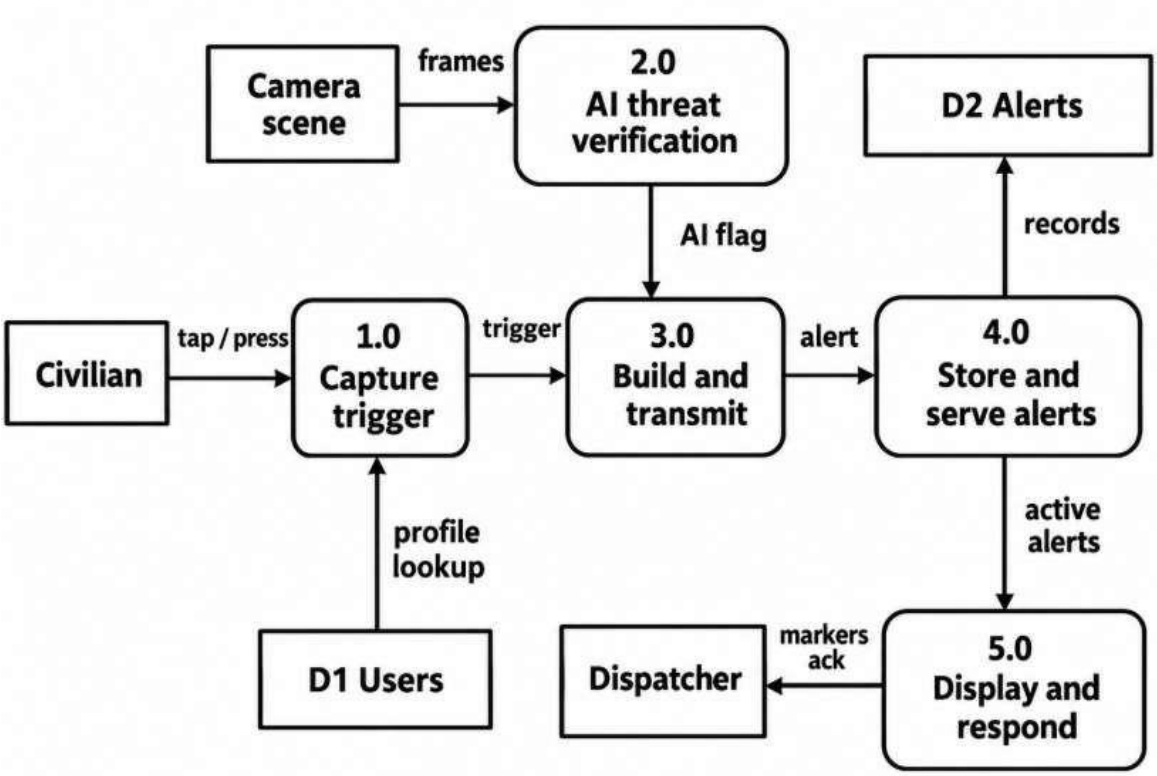


Fig.3.7Level-1DataFlowDiagram oftheSystem

THREATMODELANDATTACKSURFACE

Several specific threats were considered during design. Alert spoofing is mitigated through per-device authentication tokens verified by the Flask API. RFID cards are inherently easy to clone, so the card is treated only as a hint of identity rather than a full authentication factor, with UIDs cross-checked server-side an attacker in possession of a cloned card can trigger an alert but cannot assume a verified identity. WiFi routers were assumedlikelytofailorfacejamming,whichmotivatedtheLoRaandGSMfallbackpaths as an availability control rather than purely a reliability feature. The camera and its inferencepipelinerepresentafurtherattacksurface:afalse-positive-inducingobjectcould beusedtotriggernuisancealerts,whichismitigatedbykeepingahumandispatcherinthe loop rather than allowing the system to auto-dispatch a response.

Table3.3 lists the threats identified during thedesign review, thepossible impact of each, and the mitigation adopted. The identifiers T1 to T7 are also marked on the attack-surface view in Fig. 3.8, which maps each threat to the component or link it targets.

TABLE3.2THREATANALYSISAND MITIGATION

ID	Threat	Impact	Mitigation
T1	Alert spoofing throughforgedcalls to the Flask API	Falseincidents;wasted patrol effort	Per-deviceauthenticationtokensverified bytheFlaskAPI;Firebasesecurityrules
T2	RFIDcardcloning	Alertraisedunderanother person's identity	Card treated as a hint of identity, not an authenticationfactor;UIDscross-checked server-side
T3	WiFirouterfailureor jamming	Primarychannellost	LoRaandGSMfallbackpathsactasan availability control
T4	Look-alikeobjects used to trigger nuisance AI alerts	Alertfatiguefor dispatchers	Humandispatcherreviewseveryalert;no automatic dispatch
T5	Eavesdropping on LoRaorGSMtraffic	Exposureofdevice location	Minimalpayload(deviceID,coordinates, alert type); details resolved server-side
T6	Unauthorizedaccess to the dashboard	Disclosureofincidentsand profiles	Firebaserulesrestrictaccessto authenticated dispatchers only
T7	Over-retention of footageandprofile data	Privacyintrusion	Strictretentionlimits,signage,and minimal consent-based profiles

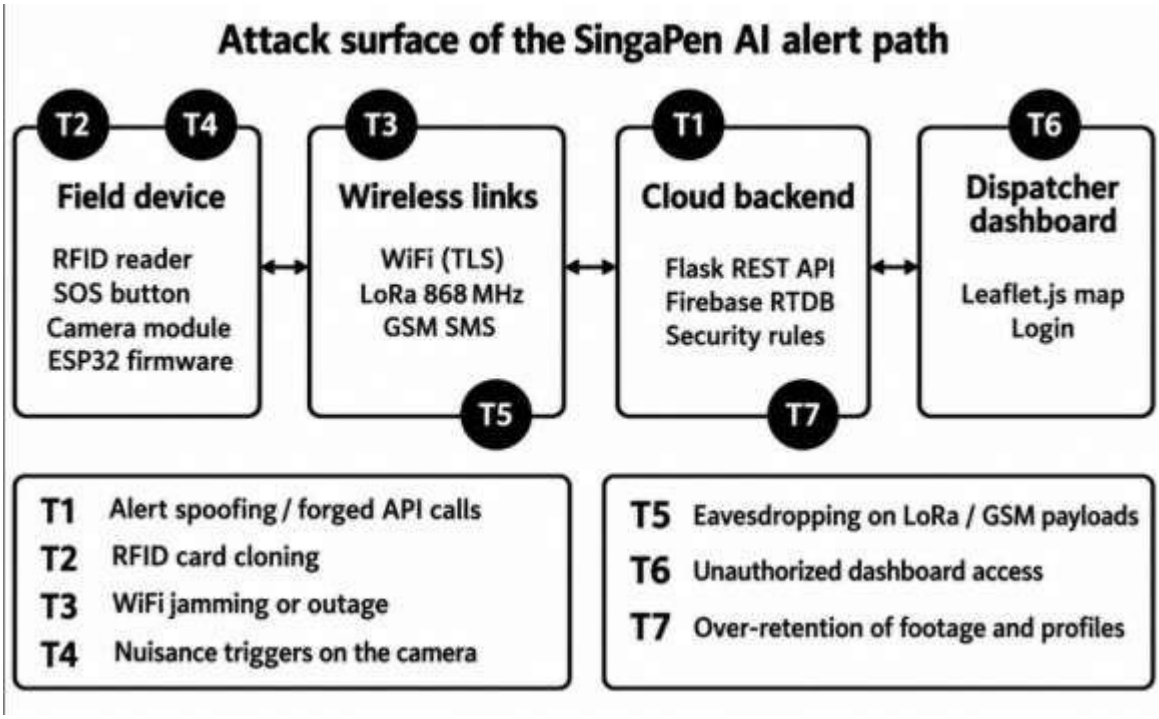


Fig.3.8AttackSurfaceoftheAlertPathwithThreatMarkersT1–T7

SECURITYCONTROLS

The WiFi link to Firebase relies on standard TLS-secured HTTPS, offering a well-establishedbalanceofsafetyandrobustness.TheLoRaconnectionismanageddifferently: transmissiontakesplaceoverthe868MHzISMfrequencyusingtheSX1278module,and due to LoRa bandwidth constraints, a streamlined custom framing layer was created that sendsonlythedeviceID,GPSlocation,andaconcisealert-typeidentifier.ThroughGSM, the ESP32 transmits standard AT commands via UART to the SIM800L module, dispatching preformatted SMS messages to saved contacts.

TrafficcrossingtheWiFinetworkisencryptedunderTLS1.2orlaterviatheFirebase SDK. Stored alerts and profiles in the cloud are protected by Firebase security rules, restrictingaccesstoauthenticateddispatchersonly.BecauseLoRaandGSMoperateunder tight bandwidth constraints, the full payload is not encrypted on those channels; instead, onlyabaredeviceIDandcoordinatepairistransmitted,withtheserverresolvingcomplete incident details internally once that initial signal arrives.

Thecontrolsappliedtoeachchanneldifferbecausethechannelsdifferinbandwidth and in the trust that can be placed in them. Table 3.3 summarizes the protection applied, the content carried, and the residual risk that remains on each path.

TABLE3.3SECURITYCONTROLSBYCOMMUNICATIONCHANNEL

Channel	Protectionapplied	Contentcarried	Residualrisk
WiFi / Firebase	TLS 1.2 or later via the Firebase SDK; Firebase securityrules;per-device token	Fullalertrecord:device ID, GPS, timestamp, profile reference,alerttype,AIflag	Dependsonrouterand backhaul; susceptible to jamming
LoRa 868 MHz	Streamlined custom framing; minimalpayload;full-payload encryptionomittedbecauseof bandwidth limits	DeviceID,GPSlocation, alert-type identifier	Interception reveals little;single-hoprange is limited
GSMSMS	ATcommandsoverUARTto the SIM800L; preformatted messages to saved contacts	Preformattedmessagewith device ID and coordinates	Carrierlatencyof6–8 s; SMS is not end-to-end encrypted

LAYEREDVIEWOFTHEARCHITECTURE

ThearchitectureofSection3.1canalsobereadasfivelayers,showninFig.3.9.The perception layer contains the sensors that capture identity, manual triggers, location, and imagery. The edge-processing layer, built around the ESP32, runs the event loop, the YOLO11n inference, and the payload builder. The communication layer offers three independent paths to the cloud. The cloud layer receives alerts through the Flask API and the Firebase Realtime Database, and the application layer presents them to the dispatcher. Separating the layers in this way allows each to be improved independently; for example, thevision model can bereplaced without changing theradio layer, and anewchannel can be added without modifying the dashboard.

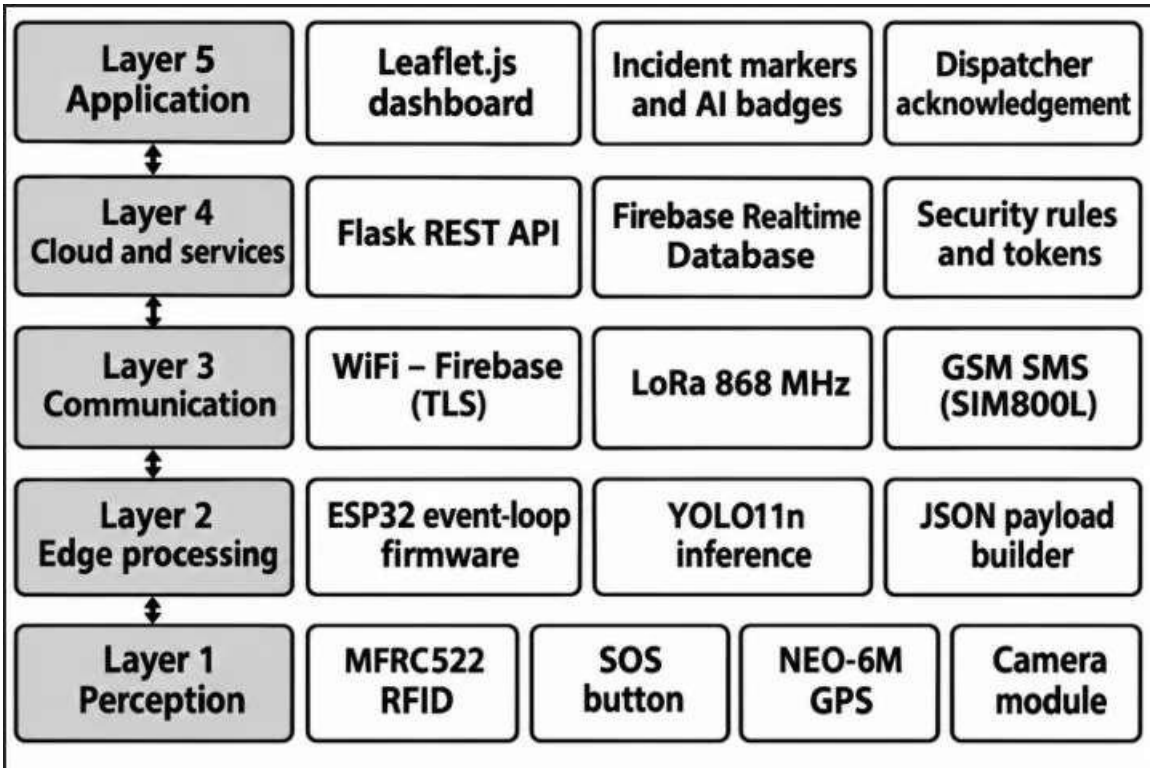


Fig.3.9LayeredViewoftheSingaPenAIArchitecture

ALERTDATAMODEL

Both hardware forms share one alert schema, which is what allows a fixed pole and aportablewearabletoappearonthesamedashboard.Table3.4liststhelogicalfieldsofan alert record and the channels that carry them. The full record travels over WiFi/Firebase, whereas LoRa and GSM carry only the minimal fields, and the server resolves the remaining details once the initial signal arrives.

TABLE3.4LOGICALFIELDSOFTHEALERTRECORD

Field	Description	Carriedover
Device ID	Identifiesthepoleorportableunitthatraisedthe alert	WiFi,LoRa, GSM
GPSlocation	LatitudeandlongitudefromtheNEO-6Mreceiver	WiFi,LoRa, GSM
Alerttype	Triggersource:RFID,SOSbutton,orAIdetection	WiFi,LoRa, GSM
Timestamp	Timeatwhichthealertwasconstructed	WiFi/Firebase
Userprofile	RFID-linkedprofilethathelpsdispatchersidentify who needs assistance	WiFi/Firebase;resolved server-sideforLoRaand GSM
AIverificationflag	Weapon-classdetectionwithboundingboxand confidence	WiFi/Firebase
Status	Active,acknowledged,orresolved;updatedbythe dispatcher	Cloud only

TRACEABILITYOFOBJECTIVESTODESIGN

To confirm that the design addresses each objective listed in Section 1.3, Table 3.5 traces every objective to the design features that realize it and to the place where the supporting evidence is presented.

TABLE3.5TRACEABILITYOFPROJECTOBJECTIVESTODESIGN FEATURES

Objective	Designfeature	Evidence
Lamp-postemergencypole withRFID,SOS,GPS,and YOLO verification	ESP32controllerwithMFRC522,NEO-6M, camera module, and YOLO11n (Table 3.1)	Figs.3.1and3.2;Section 4.2.1
PortableSOSdevicesharing the identical alert schema	PortableGuardianbuiltontheESP32 with GSM and LoRa (Table 3.2)	Section4.1;Table 3.5
Three-channeltransmissionthat degrades gracefully	ParallelWiFi/Firebase,LoRa,andGSM transmission	Fig.3.6;Table3.4
Flask-baseddashboardmapping incidents in real time	FlaskRESTAPI,FirebaseRealtime Database, Leaflet.js dashboard	Section3.2;Fig. 4.6
Empiricalevaluationoflatency, reliability, and AI accuracy	Campusdeploymentwithdefinedtest scenarios	Tables4.1and4.3; Figs. 4.9and4.10

CHAPTER 4

SYSTEMDESCRIPTION

DATACOLLECTION

Mode1–SmartRFIDEmergencyPole:thepolespendsmostofitsoperatingtimein alow-powerstandbystate,pollingtheRFIDreaderevery50mswhilemonitoringtheSOS button throughaGPIOinterrupt. When an RFID cardis tapped, theRC522 reads the UID in approximately 35 ms, and the ESP32 immediately queries Firebase for the associated profile.ApressoftheSOSbuttonisdebouncedover100mstoruleoutaccidentaltriggers. The NEO-6M module then acquires a GPS fix, typically under 5 seconds with location assistance, though a cold start can take up to 28 seconds. Once coordinates are available, the device ID, location, timestamp, user profile, and alert type are packaged into a JSON payload and transmitted simultaneously across LoRa (868 MHz), GSM SMS, and WiFi/Firebase. Ifthecameraisactive, YOLO11nanalyzesthecurrentframeforweapons.

Mode 2 – Portable Guardian SOS Device: the portable device follows a simple operating cycle boot, acquire GPS coordinates, and enter deep-sleep standby. A 3-second press of the SOS button wakes the device and constructs an alert, using the most recently cachedGPSfixtosavetimebeforetransmittingoverGSMandLoRa.Toassistresponders in tracking the user afterward, GPS coordinates are refreshed every 30 seconds for five minutes following an alert. User feedback is kept minimal: an OLED display confirms “ALERT SENT,” and a buzzer emits three short beeps.

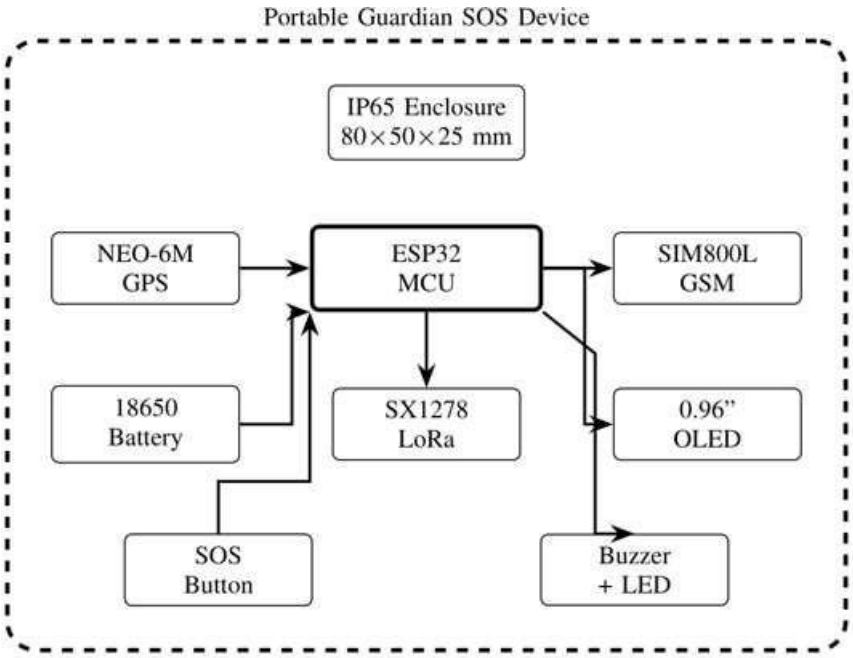


Fig.4.1PortableGuardianSOS Device

Three priorities guided the Portable Guardian design: simplicity of use, communicationreliability,andaffordability.AsingleSOSbuttonwaschosendeliberately to minimize cognitive load during high-stress moments, making the device usable regardless of age or technical familiarity. Redundant wireless channels (GSM, LoRa, and WiFi) ensure the alert still reaches the cloud even when one network path fails entirely. Physical dimensions settled at 80 mm × 50 mm × 25 mm and a weight of approximately 120 g. The device runs on a 3.7 V, 2600 mAh Li-Ion battery, providing roughly 48 hours of standby or 10 hours of active use, and requires about 3 hours to fully charge via the TP4056. GPS accuracy is approximately 2.5 m in autonomous mode, the SOS trigger requiresadeliberate3-secondlongpress,andtheunitisratedforoperationbetween10°C and +60°C within an IP65 enclosure.

INSTALLATIONLAYOUTOFTHESMARTRFIDEMERGENCYPOLE

Fig.4.2showshowthecomponentsofthefixed unitarearranged on alamppost. In this layout the camera is mounted high on the pole for a wide field of view, whereas the RFIDreader,SOSbutton,andspeakerandmicrophoneareplacedataheightthataperson canreacheasily.TheESP32,NEO-6M,SIM800L,SX1278,andbuckconverterarehoused together in a single enclosure, and the buck converter regulates the incoming supply to consistent5Vand3.3Vrails.TheantennasforGPS,LoRa,andGSMarekeptoutsidethe enclosure so that radio performance is not degraded.

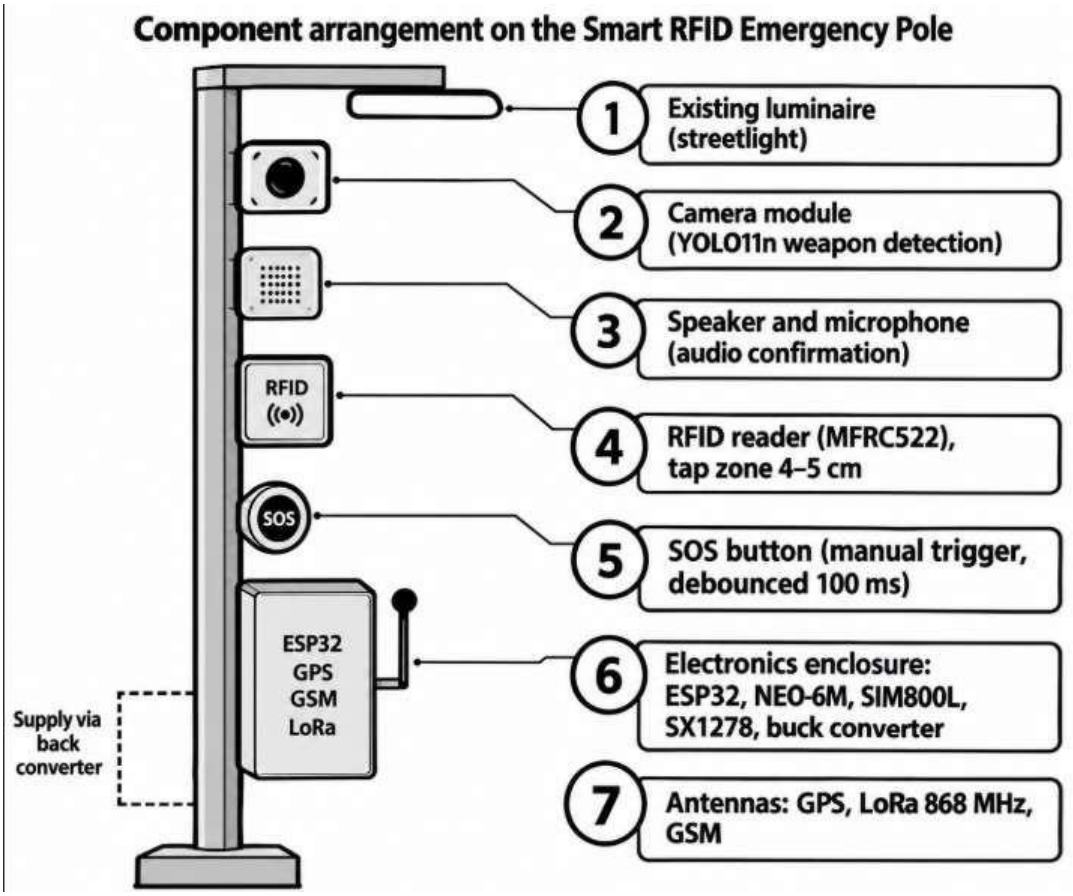


Fig.4.2ComponentArrangementontheSmartRFIDEmergencyPole (Schematic)

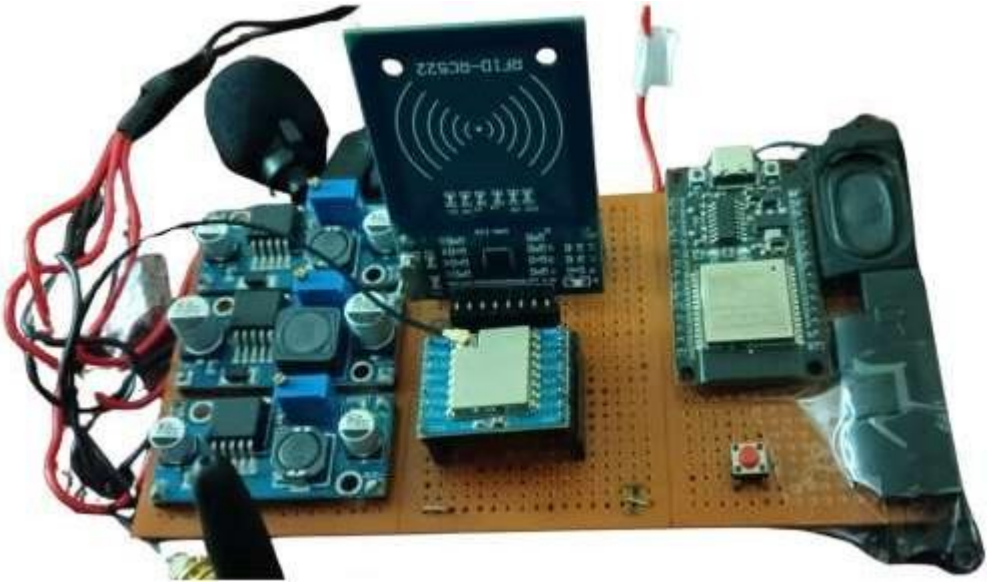


Fig.4.3FrontviewofsigapenAIprototype device



Fig4.4BacksideviewofSingapenAI prototype

ACQUISITIONANDTIMINGPARAMETERS

Table 4.2 consolidates the timing and operating parameters that govern data collection on the two hardware forms. These are the values used during the campus evaluation, and they explain the responsiveness figures reported in Section 4.2.2.

TABLE4.2ACQUISITIONANDOPERATING PARAMETERS

Parameter	Value	Appliesto
RFIDpollinginterval	50 ms	Pole
RFIDUIDread time	≈35 ms	Pole
SOSdebounceperiod	100 ms	Pole
SOSlong-pressduration	3 s	Portabledevice
GPSfix,assisted	<5 s	Poleandportable device
GPScoldstart	Upto28 s	Poleandportable device
GPSrefreshafteranalert	Every30sfor5min	Portabledevice
AIconfidencethreshold (τ)	0.20	Polecamera
Dashboardpollinginterval	1 s	Dashboard
Battery	3.7V,2600mAhLi-Ion	Portabledevice
Enclosure,size,and mass	IP65,80×50×25mm,≈120g	Portabledevice

DATAPREPROCESSING

ALGORITHMS/METHODOLOGY

The vision pipeline is built around Ultralytics YOLO11n, a compact single-stage detector selected specifically for its ability to run efficiently on edge-adjacent hardware. Frames captured by the pole camera undergo minimal preprocessing mainly resizing and normalization — prior to being processed by the network, which outputs bounding boxes andcorrespondingconfidencescores.DetectionislimitedtotheCOCOweaponcategories thatincludeknives(class43)andscissors(class76).Anydetectionwithconfidenceabove 0.20, thethreshold set forthis prototype,sets anAI-corroboratedalert flag that is attached to the outgoing JSON payload. Notably, the camera can independently initiate an alert based purely on AI detection even when the SOS button is never pressed.

Pole-sidealertconstruction(Algorithm1):onanRFIDtap,theUIDisreadwitha35 ms timeout and looked up in Firebase; on an SOS press, the input is debounced over 100 ms.GPSisthenacquired(assistedfix<5s,coldstart≤28s),aJSONpayloadisbuiltfrom thedeviceID,GPS,timestamp,andprofile,thecameraframeisrunthroughYOLO

inference if active, and the payload is transmitted over LoRa, GSM, and Firebase in sequence.

AI weapon-class verification (Algorithm 2): each frame is preprocessed and passed throughYOLO11ninference;foreverydetectionwhoseclassisknifeorscissorsandwhose confidenceisatleastthethresholdτ=0.20,thedetection,itsboundingbox,andconfidence are returned as a positive AI verification.

Two mathematical models support the system. The great-circle distance between an alert and the nearest patrol unit is computed with the Haversine formula, using angular differences $\Delta\varphi$ and $\Delta\lambda$ between the alert location ($\varphi1$, $\lambda1$) and the responder location ($\varphi2$, $\lambda2$), with Earth radius $R \approx 6{,}371$ km. Detection performance is evaluated with standard metricsrestrictedtotheweaponclassespresentinthetestdata:Precision,Recall,F1-score, Intersection over Union (IoU), and mean Average Precision (mAP). End-to-end alert latency L is defined as the cumulative delay across trigger, GPS acquisition, payload construction, transmission, cloud update, and dashboard rendering, while radio reliability is assessed via packet delivery ratio (PDR) for both LoRa and GSM, and portable-device batterylifeis estimatedfromaduty-cycledformulausingactive/standbycurrentdrawand capacity.

VisionPipeline

The vision pipeline is summarized in Fig. 4.5. Each frame from the pole camera is resized and normalized to 640 ×640 pixels and passedthroughYOLO11n, whoseoutputs are filtered to the knife and scissors classes. A detection is accepted only if its confidence reaches the threshold $\tau = 0.20$. An accepted detection sets the AI-corroborated flag and attachestheboundingboxtotheoutgoingpayload,whileanydetectionbelowthethreshold is discarded. The deliberately low threshold favors recall, since the dispatcher reviews every flagged frame and a missed weapon is costlier than a nuisance flag.

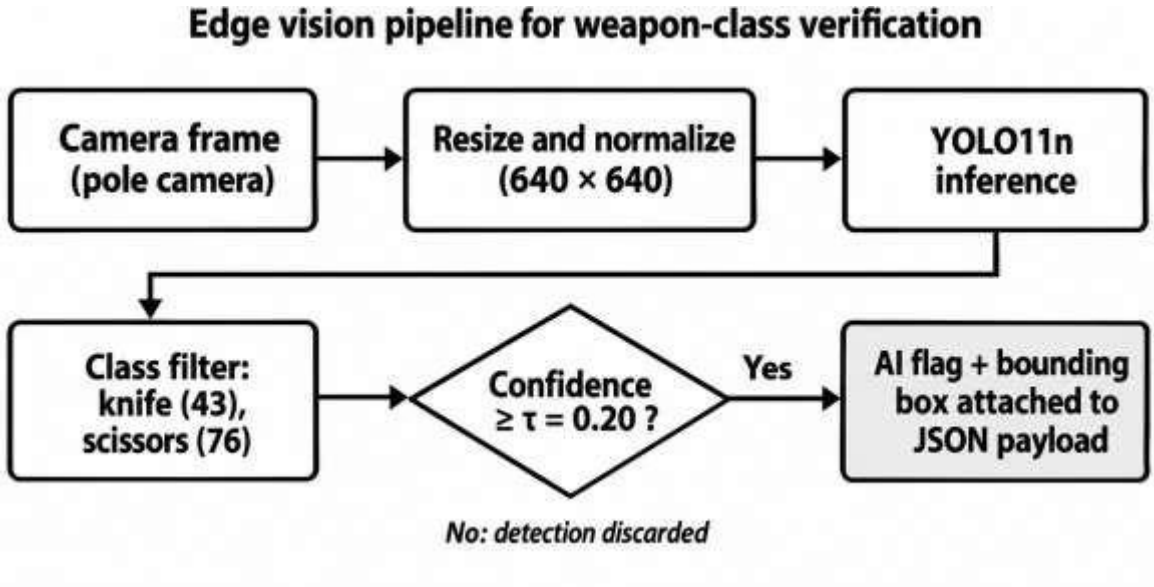


Fig.4.5EdgeVisionPipelineforWeapon-Class Verification

MathematicalFormulation

The quantities used to design and evaluate the system are defined below. The great-circle distance d between an alert location (φ_1, λ_1) and a responder location (φ_2, λ_2) is obtained with the Haversine formula, where $\Delta\varphi$ and $\Delta\lambda$ are the angular differences in latitude and longitude and $R \approx 6{,}371$ km is the Earth radius:

$$a=\sin^2(\Delta\varphi/2) +\cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda / 2) \quad (1)$$

$$d=2R \cdot \operatorname{atan2}(\sqrt{a},\sqrt{(1 - a)}) \quad (2)$$

Detection quality is measured with precision, recall, and F1-score, and with the intersectionoverunion(IoU)betweenapredictedbox$B_p$andaground-truthboxB_g.Mean average precision (mAP) is the mean of the per-class average precision values, restricted to the weapon classes present in the test data:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \quad (3)$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (4)$$

$$IoU = |B_p \cap B_g| / |B_p \cup B_g| \quad (5)$$

End-to-end alert latency L is the sum of the trigger, GPS acquisition, payload construction, transmission, cloud update, and dashboard rendering delays, and the packet delivery ratio (PDR) is the percentage of transmitted packets that are received:

$$L=t_{trigger}+t_{GPS}+t_{build}+t_{tx}+t_{cloud}+t_{dash} \quad (6)$$

$$PDR=(N_{received}/N_{sent})\times 100\% \quad (7)$$

For the portable device, the average current I_{avg}follows from the duty cycle D of active operation, and the battery life T_{batt}follows from the rated capacity C and a usable-capacity factor η:

$$I_{avg}= D \cdot I_{active}+ (1 - D) \cdot I_{sleep} \quad (8)$$

$$T_{batt}=(C \cdot \eta)/ I_{avg} \quad (9)$$

Taking C = 2,600 mAh, an average active draw of 228 mA, and $\eta \approx 0.85$ gives approximately 9.7 hours, which agrees with the active-mode battery lifereported in Table 4.1.

DispatcherDashboard

The dispatcher dashboard, illustrated schematically in Fig. 4.6, is a single-page Leaflet.jsapplicationthatreadsthealertfeedfromFirebaseataone-secondinterval.Each active incident appears as a marker on the map, with distinct symbols for pole-raised and portable-device alerts, an AI badge where the vision pipeline has flagged a weapon class, and a camera thumbnail in the side panel. The dispatcher acknowledges an alert through thePOST/api/alert/{id}/ackendpointandmarksitresolvedoncetheresponseiscomplete. Becausetheinterfacepresentstheevidencebutleavesthedecisiontoaperson,thesystem operates as a triage aid and never dispatches a patrol automatically.

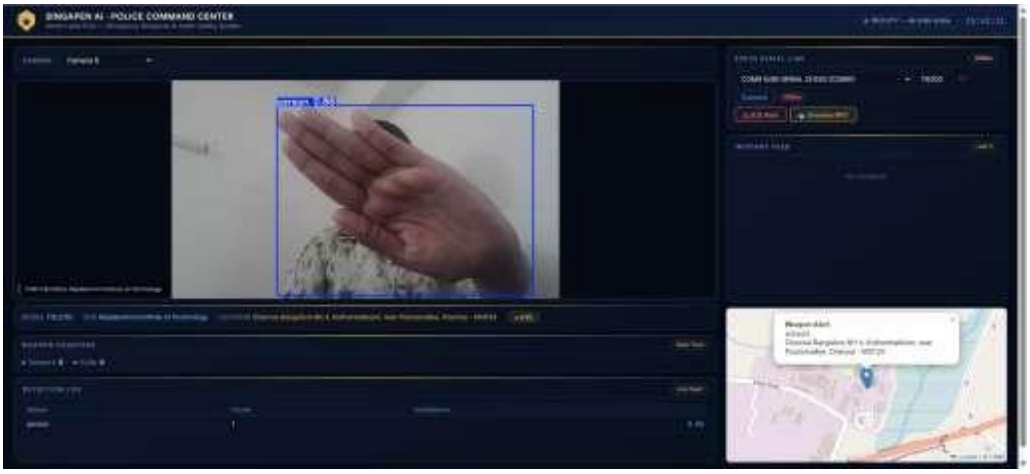


Fig.4.6SchematicLayoutoftheDispatcher Dashboard



Fig4.7AIbasedweapondetectionalerttriggerautomaticallywith location



Fig4.8SOSAlerttriggeranddashboardindicatorwith location

RESULTSANDDISCUSSION

Pole firmware was written in C++ using the Arduino/ESP-IDF toolchain. RFID and LoRa peripherals were connected over SPI, while GPS and GSM modules used standard UART.Thebackend was implementedin Pythonusing FlaskforRESTendpoints and the FirebaseAdminSDKfordatabasesynchronization.TheAIpipelinecentersonUltralytics YOLO11n, with OpenCV capturing and preprocessing frames. Testing was conducted on both a CPU-only development machine and a GPU-equipped workstation to capture the fullrange ofinferencelatency,and asingle-page Leaflet.jsdashboardpolledFirebase at a 1-secondinterval.Real-worldtestingtookplacethroughacampusdeploymentat

RajalakshmiEngineeringCollege,Chennai,withLoRarangetestedat1kmand5kmline of sight, GSM delivery tested across two carriers, and AI accuracy probed with a labeled set of knife and scissors images mixed with visually similar distractor objects.

TABLE4.1KEYPERFORMANCEMETRICS

Metric	**Value**	**Notes**
AIInference(CPU)	45 ms/frame	YOLO11n,640×640
AIInference(GPU)	2.5 ms/frame	-
GPSAccuracy(CEP)	2.8 m	NEO-6M,8–10sats
GPSCold Start	28 s	Assisted:<5s
RFIDRead Range	4–5 cm	100%UID match
RFIDReadTime	35 msavg	-
LoRa1km LOS	100%	Packetsuccess
LoRa5km LOS	85%	Packetsuccess
GSMSMS(CarrierA)	90% delivery	8 s average
GSMSMS(CarrierB)	100% delivery	6 s average
EndtoEnd(WiFi)	803 ms	-
EndtoEnd(LoRa)	1,253ms	-
EndtoEnd(GSM)	7,303ms	-
DashboardLatency	< 1 s	1s polling
Battery(Active)	9.7 h	228mAavg draw
Battery(Standby)	> 48 h	Deepsleep5µA

TheWiFi/FirebasepathprovedtobethefastestchannelbyaclearmargIn,averaging about 803 ms, making it well suited as the primary route. LoRa required a bit more time, providing alerts at around 1,253 ms on average, yet served as a dependable, cellular-independentalternative.GSMSMSexperiencedsignificantdelays,averagingbetween6to

8secondsbasedonthecarrier;thoughnotidealasamainchannel,itstillservesasareliable alternative for alerting emergency contacts when other options are inaccessible.

On the AI side, the system accurately detected scissors and knives at a 0.20 confidencethreshold,withafalsepositiveratearound5percent,mainlycausedbyvisually similar items like pliers and letter openers. This error rate was viewed as acceptable in practice,sinceahumandispatcheralwaysexaminesthefootagebeforesendingoutapatrol. GPS accuracy measured 2.8 m CEP, well within the precision needed for emergency dispatch, and the dashboard sustained its 1-second polling interval without degradation even under continuous video streaming.

Severallimitationsemergedduringtesting.The6to8secondGSMdeliverylatency rules it out as a sole primary channel, so it functions strictly as a backup. LoRa packet successalsodroppedfromaperfect100percentat1kmto85percentat5kmlineofsight, suggestingthatlongercorridorswouldrequiretighternodespacingorapproperrelay/mesh topology. The AI's 5 percent false-positive rate confirms that the system functions as a triagetoolratherthananautonomousdecision-maker,andtheGPScold-startlatencyofup to28secondscoulddelayafixifthedevicewerefullypoweredoff,partiallyaddressedby waking periodically for GPS updates during standby.

GRAPHICALANALYSISOFRESULTS

The measurements in Table 4.1 are presented graphically in Fig. 4.9 and Fig. 4.10. Fig. 4.9 compares end-to-end latency on thethree channels. WiFi/Firebase is thefastest at 803ms,LoRaisabout1.6timesslowerat1,253ms,andGSMSMSat7,303msisroughly ninetimesslowerthanWiFi.Therankingmatchestheintendedrolesofthechannels:WiFi as the primary route, LoRa as the cellular-independent alternative, and GSM as the last-resort notification path.

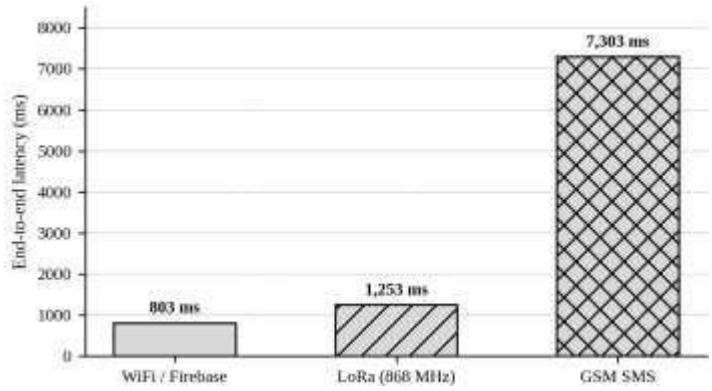


Fig.4.9End-to-EndAlertLatencybyChannel

Fig.4.10showsdeliverysuccess.LoRadeliveredeverypacketat1kmand85percent at 5 km in line of sight, while SMS delivery was 90 percent on Carrier A and 100 percent onCarrierB.Thefall-offofLoRawithdistanceandthecarrier-to-carriervariationinSMS deliverybothargueforkeepingallthreechannelsactiveatonceratherthanrelyingonany singleone.OntheAIside,YOLO11ninferencetook45msper640×640frameonaCPU-only machine and 2.5 ms on a GPU (Table 4.1), so even CPU-only hardware can process more than 20 frames per second.

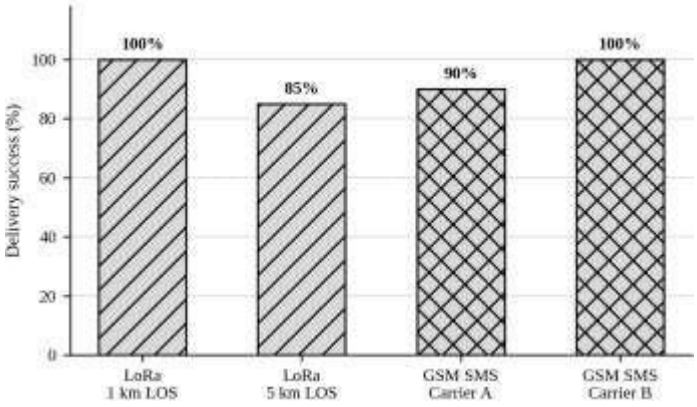


Fig.4.10DeliverySuccessoftheLoRaand GSM Channels

TESTSCENARIOSANDOBSERVEDOUTCOMES

The campus evaluation covered the ten scenarios listed in Table 4.3. For each scenariothetablerecordstheconditionsunderwhichitwasrunandtheoutcomeobserved, using the same measurements that appear in Table 4.1.

TABLE4.3TESTSCENARIOSANDOBSERVED OUTCOMES

ID	Scenario	Conditions	Observedoutcome
S1	RFIDidentitytrigger	Cardtappedwithinread range	100%UIDmatchat4–5cm;35ms average read time
S2	ManualSOStrigger	Buttonpresswith100ms debounce	Accidentalcontactsrejected;deliberate press raises the alert
S3	WiFi/Firebasepath	CampusWiFi;dashboard polling at 1 s	803msaverageend-to-endlatency
S4	LoRalink,1km	Lineof sight	100%packetsuccess;1,253msend-to-end latency
S5	LoRalink,5km	Lineof sight	85%packet success
S6	GSMSMS,CarrierA	Livecellularnetwork	90%delivery;8 saverage
S7	GSMSMS,CarrierB	Livecellularnetwork	100%delivery;6saverage
S8	AIweapondetection	Labeled knife and scissors imagesmixedwithlook-alike objects; $\tau = 0.20$	Knives and scissors detected; false positiverateabout5%(pliers,letter openers)
S9	GPSaccuracyand start-up	8–10satellitesinview	2.8mCEP;assistedfixunder5s;cold start 28 s
S10	Dashboardunderload	Continuousvideostreaming; 1 s polling	Latencybelow1swithnodegradation

The two weakest results, LoRa delivery at 5 km and GSM latency, are discussed as limitations in Section 4.2.6.

DEPLOYMENTCOSTPROJECTION

Becausethedesigntargetsmunicipalandcampusbudgets,itisusefultoprojectwhat a corridor deployment would cost. Using the prototype figures from Appendix A.2 (Rs. 9,000 to 9,500 per pole and about Rs. 500 per pole per year for maintenance), Table 4.4 estimates the cost per kilometre for the pole spacings of 50 m and 100 m mentioned in Section 1.2. The estimate covers pole hardware and maintenance only; installation labor and civil work depend on the site, and cloud hosting is assumed to remain within the Firebase free tier.

TABLE4.4ESTIMATEDCOSTPERKILOMETREOFCORRIDOR

Polespacing	Polesperkm	Hardwarecost(Rs.)	Annualmaintenance (Rs.)
100 m	≈10	90,000–95,000	≈5,000
50 m	≈20	180,000– 190,000	≈10,000

Even the denser spacing costs less than two lakh rupees per kilometre in hardware, whichisconsistentwiththeaimofaper-unitcostthatisrealisticformunicipalandcampus budgets.Fortheportabledevice,bulkproductionof100unitsormoreisexpectedtoreduce the bill of materials from Rs. 2,125 to about Rs. 1,710 (Appendix A.2).

LIMITATIONSANDMITIGATION

Theevaluationalsoexposedanumberoflimitations,whicharesummarizedinTable 4.5togetherwiththemitigationadoptedintheprototypeorthedirectionproposedforfuture work.

TABLE4.5OBSERVEDLIMITATIONSAND MITIGATION

Limitation	Observation	Mitigationordirection
GSMlatency	6–8sdelivery,tooslowfora primary channel	UsedstrictlyasabackuptoWiFiand LoRa
LoRarange	Successfallsfrom100%at1kmto 85% at 5 km	Tighternodespacingorarelay/mesh topology (future work)
AIfalsepositives	About5%,mainlypliersandletter openers	Humandispatcherreviewseveryalert; used for triage, not autonomous decisions
GPScoldstart	Upto28swhenfullypoweredoff	PeriodicwakeupsforGPSrefresh during standby; cached fix on the portable device
Prototypescope	Single-hopLoRalink;campus-scale validation only	Municipal rollout and regulatory certificationidentifiedasfuturescope

CHAPTER 5

CONCLUSION

This work demonstrates that a lamppost-integrated, RFID- and AI-verified emergency response system can be built from simple commodity hardware, at a budget realistic for schools and local municipalities, without sacrificing responsiveness. Sub-second alert latency was achieved on the primary channel alongside reliable AI weapon corroboration.Themeasuredresults803msoverWiFi,1,253msoverLoRa,and7,303ms via GSM, together with a 100 percent LoRa success rate at 1 km and 45 ms YOLO11n inference on CPU provide reasonable confidence that this approach could scale to real campuses and city transit systems. The portable device bill of materials totals approximately Rs. 2,125 per unit for the prototypebuild, and the full pole prototype setup (including camera and enclosure) costs approximately Rs. 9,000 to 9,500, with bulk production expected to bring the portable device cost down by about 20 percent.

Several extensions are planned for future work, including Whisper-based voice triggers supporting regional languages, ArcFace-based personnel verification, and integrationwithautonomousdronedispatchforaerialincidentviews.Additionaldirections include satellite-modem backups for fully off-grid locations, predictive crime-hotspot analytics, shifting more AI inference to the edge to reduce cloud dependency, and a companion mobile application allowing citizens to view localized alerts in real time.

KEYFINDINGSANDCONTRIBUTIONS

Themainfindingsandcontributionsofthisworkcanbesummarizedas follows.

- Asinglehardwarefamily servestwoformsofdeployment:afixedlamp-postpoleand a portable wearable share one alert schema, one cloud backend, and one dashboard.
- Threeindependentchannelsgivegracefuldegradation:alertsreachedthedashboardin about 803 ms over WiFi, 1,253 ms over LoRa, and 7,303 ms over GSM, with LoRa delivering every packet at 1 km.
- On-device AIcorroboration is achievable on commodity hardware: YOLO11n runs in 45 ms per frame on a CPU with a false positive rate of about 5 percent, and a human dispatcher remains in the loop.

- The design is economical: about Rs. 2,125 per portable device and Rs. 9,000 to 9,500 per pole, with no recurring subscription fees.
- Securityandprivacyweretreatedasdesignrequirements,throughtokenauthentication, TLS,minimalpayloadsonconstrainedchannels,signage,retentionlimits,andconsent-based profiles.

ROADMAPFORFUTUREENHANCEMENTS

Theextensions listed abovedifferin effort and in dependencies. Table 5.1 therefore groups thembypurpose,expected benefit,andthemainchallenge each willface,andFig. 5.1 suggests a phasing in which the lower-risk software and network improvements come first.

TABLE5.1FUTUREENHANCEMENTS,BENEFITSANDCHALLENGES

Enhancement	Purpose	Expectedbenefit	Mainchallenge
Companionmobile application	Letcitizensview localized alerts	Widersituational awareness	Authentication and alert-fatiguecontrol
Moreinferenceatthe edge	Reducecloud dependency	Fasterandmoreprivate verification	Limitedcomputeonthe pole controller
LoRameshandrelay nodes	Extendrangebeyonda single hop	Betterdeliverybeyond1 km	Nodeplacementand power planning
Whisper-basedvoice triggers	Hands-freealertingin regional languages	Workswhenabutton cannot be reached	Noisyoutdooraudio; false triggers
ArcFacepersonnel verification	Verifytheidentityof authorizedpersonnel	Strongeridentity assurance	Privacyandconsent requirements
Predictivecrime-hotspotanalytics	Guidepatrolplanning	Proactiveratherthan reactive safety	Dataqualityand bias
Dronedispatch	Aerialviewofan incident	Fastersituational assessment	Airspaceregulationand reliability
Satellite-modem backup	Connectivityatoff-grid locations	Alerting where no cellularorWiFiexists	Modemcostandpower draw

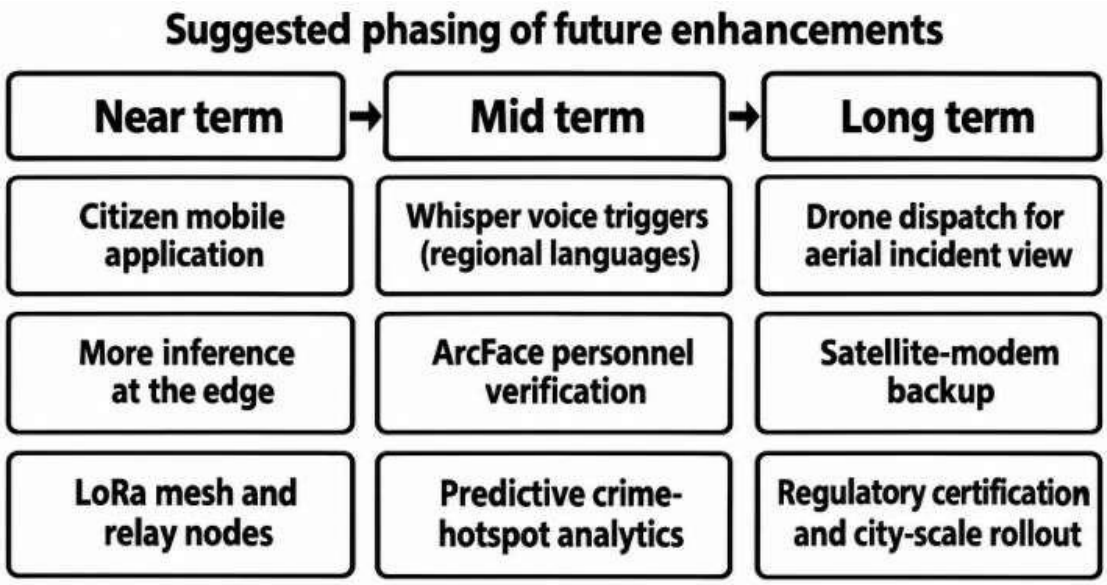


Fig.5.1SuggestedPhasingofFuture Enhancements

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APPENDIX

ApplicationsandImpact

Several practical applications follow naturally from this system. The most direct is women's safety, whether through individually carried portable devices or fixed poles installed in isolated areas that route alerts directly to the nearest all-women police station. The system also fits well on college campuses — RFID access control and AI perimeter monitoringwerepilotedatRajalakshmiInstituteofTechnologyaspartofthiswork.Transit hubs such as railway platforms and bus stops represent another strong use case, as do schools, where a simplified interface with basic geo-fencing and parent notification could be layered on top. Beyond these immediate cases, the architecture could be adapted for smart-city crime-hotspot analytics, hospital staff safety, senior-citizen monitoring, lone-worker safety in industry, and disaster management, especially since the LoRa channel operates as a relay even when the cellular network is completely offline.

CostEstimationandMaintenance

The portable device bill of materials totals approximately Rs. 2,125 per unit for the prototypebuild.Forthepolesubsystem,includingcameraandenclosure,thefullprototype setup cost approximately Rs. 9,000 to 9,500. At production volumes of100 units ormore, bulk component discounts and proper PCB integration are expected to bring the portable device cost down to roughly Rs. 1,710, a reduction of about 20 percent from prototype pricing. No recurring subscription fees were built into the design; standard GSM billing andtheFirebasefreetierareusedthroughout,and sincethefirmwareisopensource,there arenolicensingcosts.Estimatedongoingmaintenancecostsareapproximately Rs.500per pole per year, covering battery replacement and regular visual inspections.

EthicalConsiderations

Because the system tracks location and captures visual data from public spaces, ethicalsafeguardswereacentraldesignconcern.Anyreal-worlddeploymentrequiresclear signageindicatinganAI-monitoredarea.Strictdata-retentionlimitsarenecessarytoavoid indefinitely storing footage of ordinary daily activity, and dashboard access must be restrictedtoauthorizeddispatchersonly.RFID-linkedprofiledatamustalsobecollected

onlywithinformedconsent,limitedstrictlytotheminimuminformationrequiredtoassist someone during an emergency.

SampleAlertPayload

Thelistingbelowshowsanillustrativealertrecordasitmightbepushedto POST /api/alert.Fieldnamesandvaluesareexamplesonly,andthecoordinatesareplaceholders.

```
{
  "device_id":"POLE-001",
  "alert_type": "RFID",
  "timestamp":"2026-11-05T21:14:32+05:30",
  "location":{"lat":13.0000,"lon":80.0000}, "user_uid":
  "u_1024",
  "ai_verification":{
    "flag": true,
    "class": "knife",
    "confidence":0.41
  },
  "status":"active"
}
```

CommissioningChecklist

Beforeapoleorportableunitisplacedinservice,thefollowingchecksare recommended.

- **Hardware:**confirmthattheRFIDreaderreturnsthecorrectUID,thattheSOSbutton triggers after debouncing, and that the GPS module obtains a fix.
- **Connectivity:**confirmdeliveryoneachchannelseparately(WiFi/Firebase,LoRa,and GSM) and check that the dashboard shows the marker within its one-second refresh.
- **AI:**verifythatthecameraviewisunobstructedandthattheconfidencethresholdisset to the intended value.
- **Security:** confirm that the device token is provisioned, that TLS is in use on the WiFi link, and that Firebase security rules restrict access to authenticated dispatchers.
- **Privacy:**confirmthatAI-monitoringsignageisdisplayedandthatdata-retentionlimits are configured.
- **Operations:**confirmthatdispatchersaretrainedtoacknowledgeandresolvealertsand to treat AI flags as advisory.